

A Step-By-Step Guide to BirdNET Analyzer

Rebecca Cohen, Irina Tolkova, Meghan Beatty, Spencer Keyser, Isha Bopardikar, Chiti Arvind, V.V. Robin

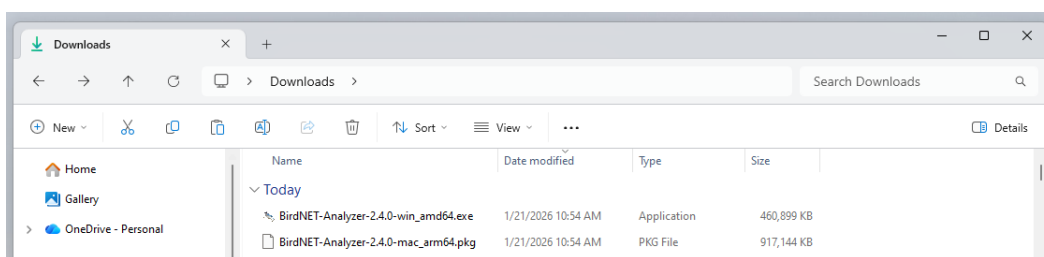
Adapted and updated from materials previously created by Ashraft Syazwan Ahmady Yusni, Nor Amira Abdul Rahman, Maria Vasilkin, and Dena Clink, and based on: Symes L, Sugai LSMS, Gottesman B, Pitzrick M, Wood C. 2023. Acoustic analysis with BirdNET and (almost) no coding: practical instructions. <https://doi.org/10.5281/zenodo.8357176>

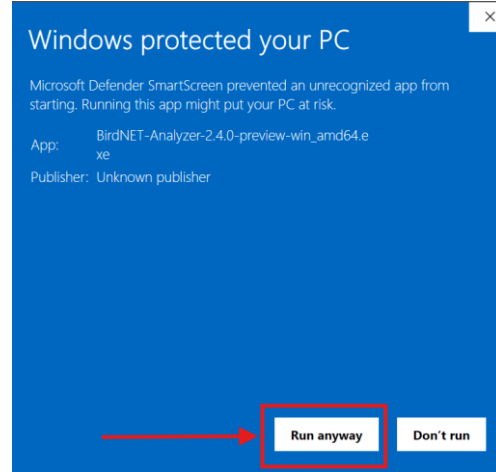
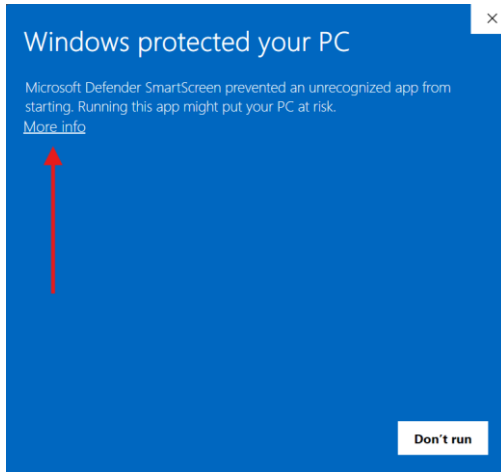
The data and downloads for this guide can be found [here](#)

Part 1: Software Install

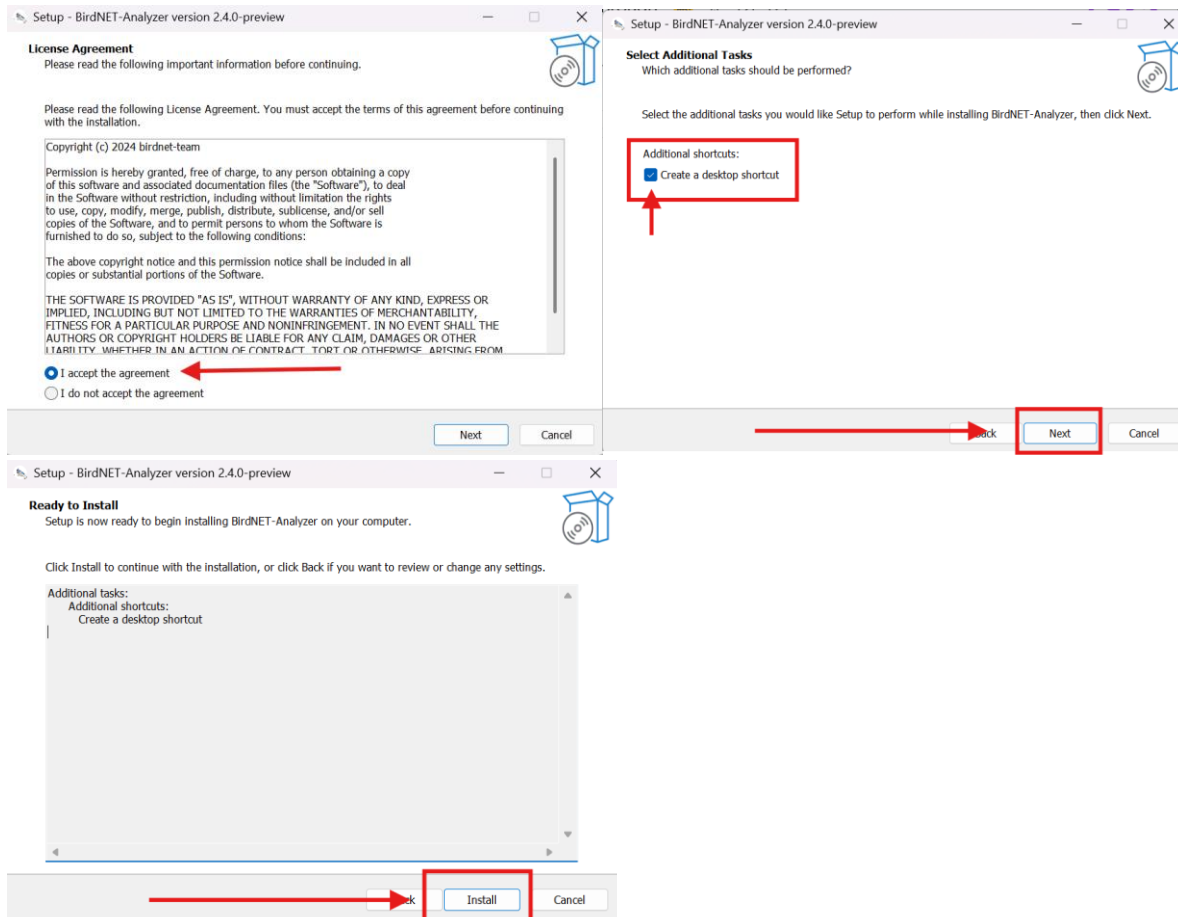
1.1 BirdNET Analyzer v.2.4.0

1. Download BirdNET Analyzer v2.4.0 installer from the [BirdNET GitHub repository](#) releases page
 - BirdNET-Analyzer-2.4.0-win_amd64.exe for Windows
 - BirdNET-Analyzer-2.4.0-mac_arm64.pkg for MAC
2. Double click the downloaded installer file to run it; you may need to tell your computer that it is safe to run. Click **More info** and **Run anyway**.





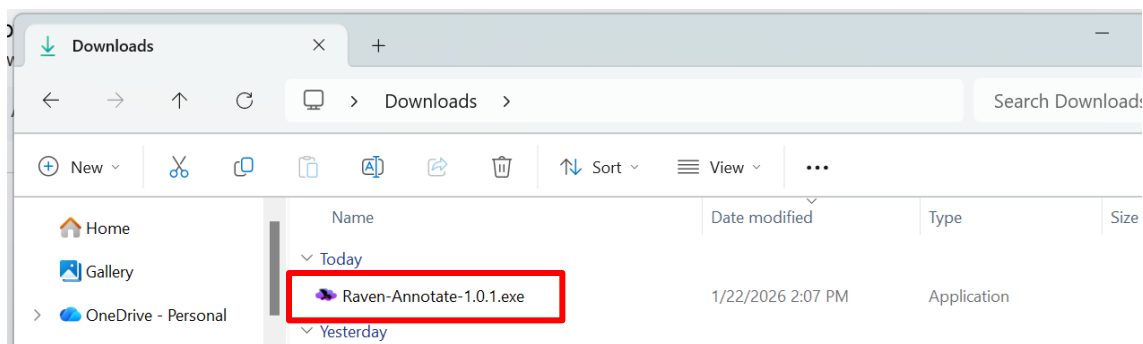
3. Accept the license agreement and click **Next**. Check the box to create a desktop shortcut and click **Next**. Click **Install**.



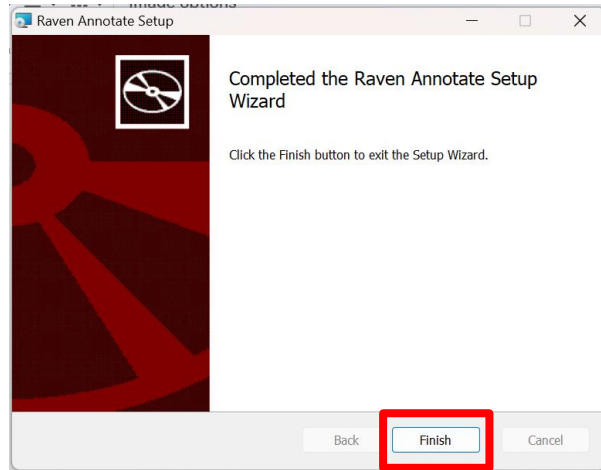
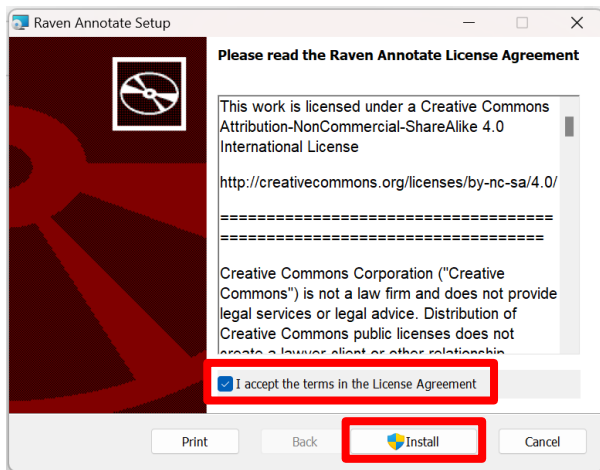
1.2 Raven Annotate Pre-Release Build (install optional)

Raven Annotate is a new Raven Workbench module that is intended to replace Raven Pro's functionality for visualizing raw audio files, annotating acoustic events, making automatic feature measurements, and more. Raven Annotate is still in development and has not yet been publicly released, but the basic visualization and annotation functionality is up and running in this build. New functionality will continue to be developed and released in coming months. **If you have Raven Pro or Raven Lite already installed, it is optional to install Raven Annotate.**

1. Use one of the following links to download the appropriate Raven Annotate installer, according to your operating system:
 - a. Windows: https://updates.ravensoundsoftware.com/updates/workbench/raven_annotate/win_x64/Raven-Annotate-1.1.0.exe
 - b. MacOS: https://updates.ravensoundsoftware.com/updates/workbench/raven_annotate/mac_arm64/Raven-Annotate-1.1.0.dmg
 - c. Linux: https://updates.ravensoundsoftware.com/updates/workbench/raven_annotate/linux_x64/raven-annotate_1.1.0-1_amd64.deb
2. Locate the installer in your Downloads folder and double click to run.



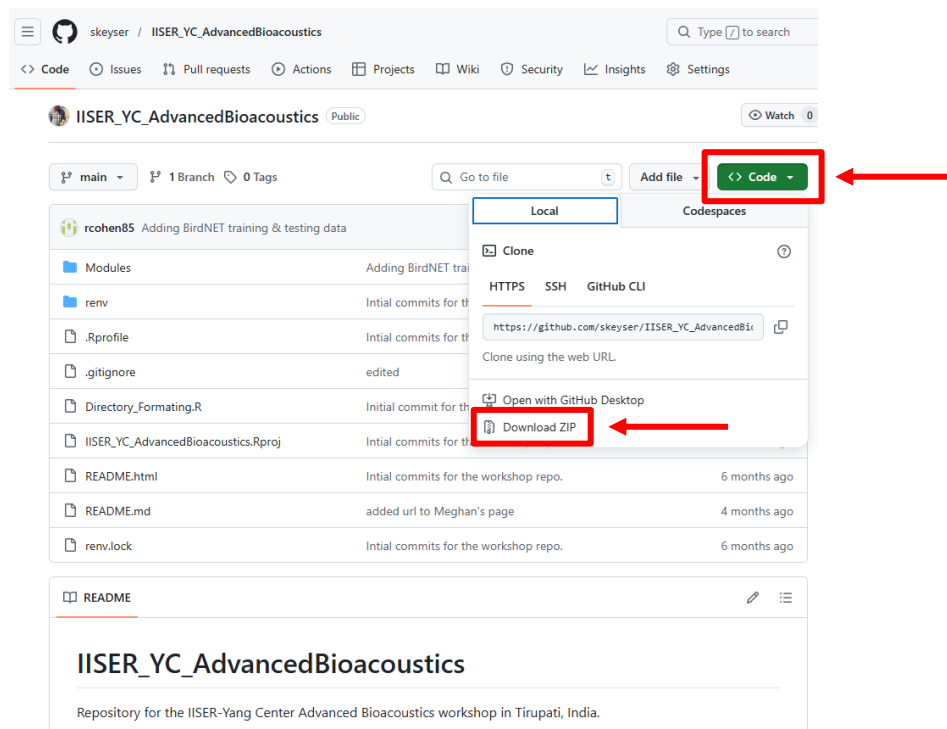
3. Follow the prompts to complete installation by accepting the license agreement and clicking ***Install***.



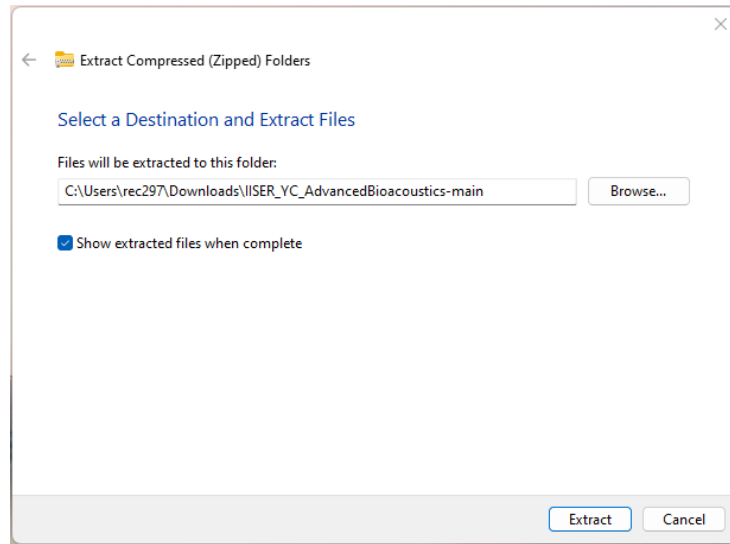
Part 2: Detecting with BirdNET Model V2.4

2.1 Download workshop materials

1. From the [GitHub](#) page, **download workshop repository** “**IISER_YC_AdvancedBioacoustics**” as a **zip file**. This repository contains all of the data and code we will be using for the modules we’ll cover this week. Slide decks will be added at the end of the workshop.



2. **Unzip the folder** and store it in a convenient location on your computer.

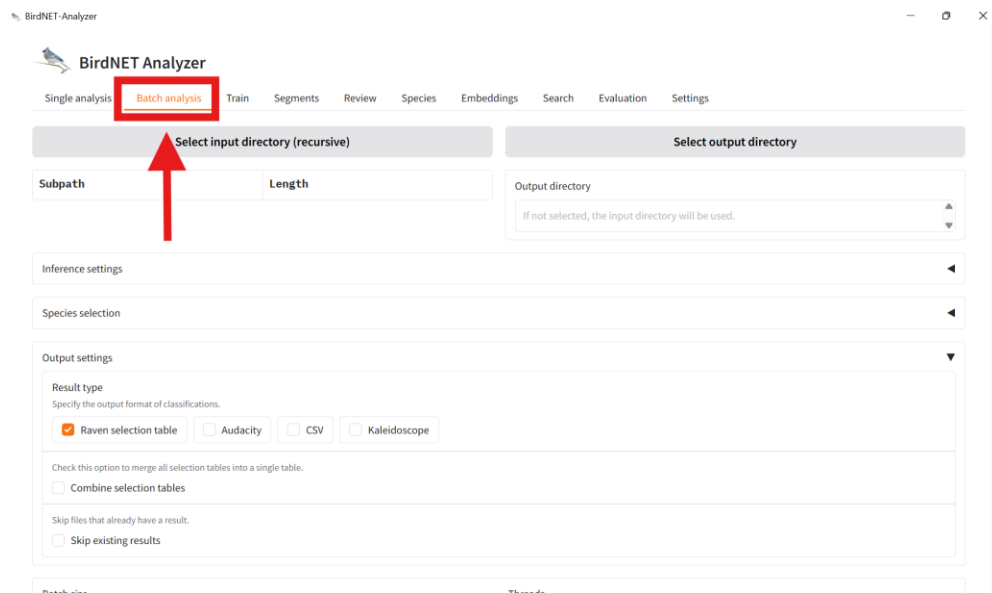


2.2 Running BirdNET Analyzer on Bird Data

1. From your Desktop, **double-click the BirdNET icon to launch BirdNET-Analyzer**. It may take a minute for the program to open.



2. Navigate to the **Batch analysis** tab.



3. Click **Select input directory** and navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing >> Audio Files >> Terrestrial*.

BirdNET Analyzer

Single analysis **Batch analysis** Train Segments Review Species Embeddings Search Evaluation Settings

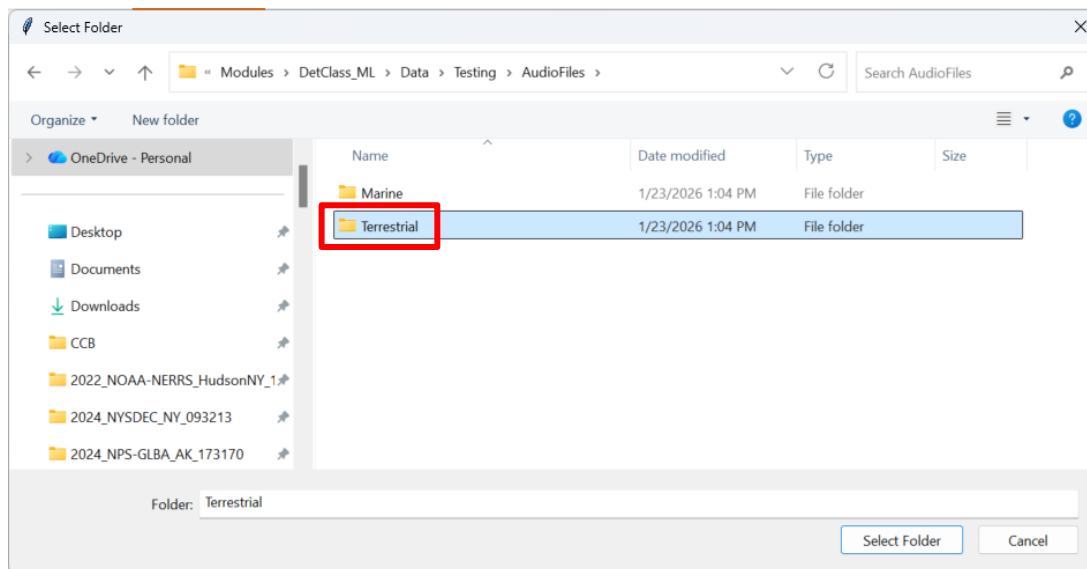
Select input directory (recursive)  Select output directory

Subpath Length Output directory
If not selected, the input directory will be used.

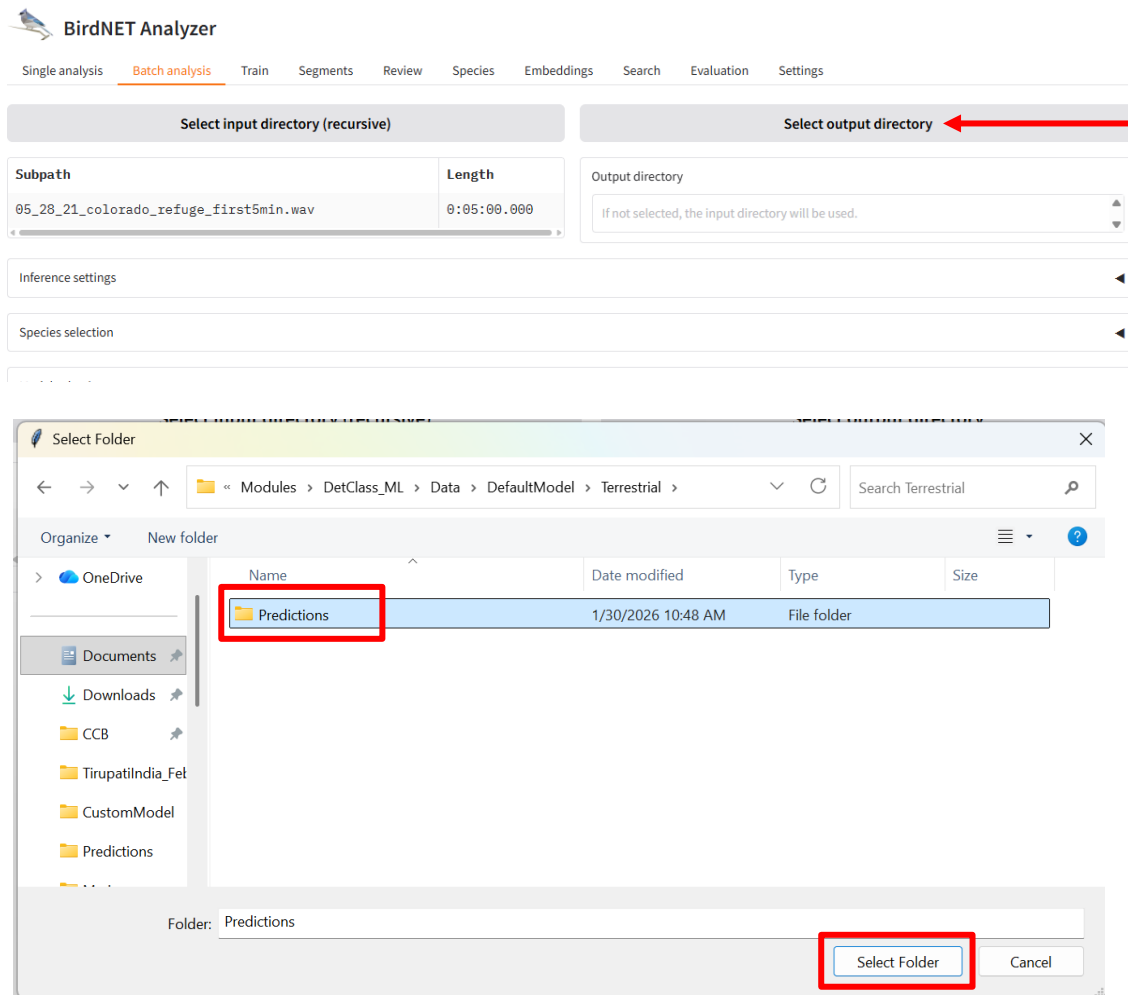
Inference settings

Species selection

Model selection



- Click **Select output directory** and navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Terrestrial >> Predictions*.



- At the bottom of the screen, click **Analyze**. A progress bar will appear. When the analysis is complete it will switch to the message "All files analyzed!" Navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Terrestrial >> Predictions* to see the output selection table.

Output settings

Result type

Specify the output format of classifications.

☒ Raven selection table

☐ Audacity

☐ CSV

☐ Kaleidoscope

Check this option to merge all selection tables into a single table.

☐ Combine selection tables

Skip files that already have a result.

☐ Skip existing results

Batch size

Number of samples to process at the same time.

1

Threads

Number of CPU threads.

4

Locale

Locale for the translated species common names in the output

EN

Analyze

Analyze

Starting ... - 0.0%

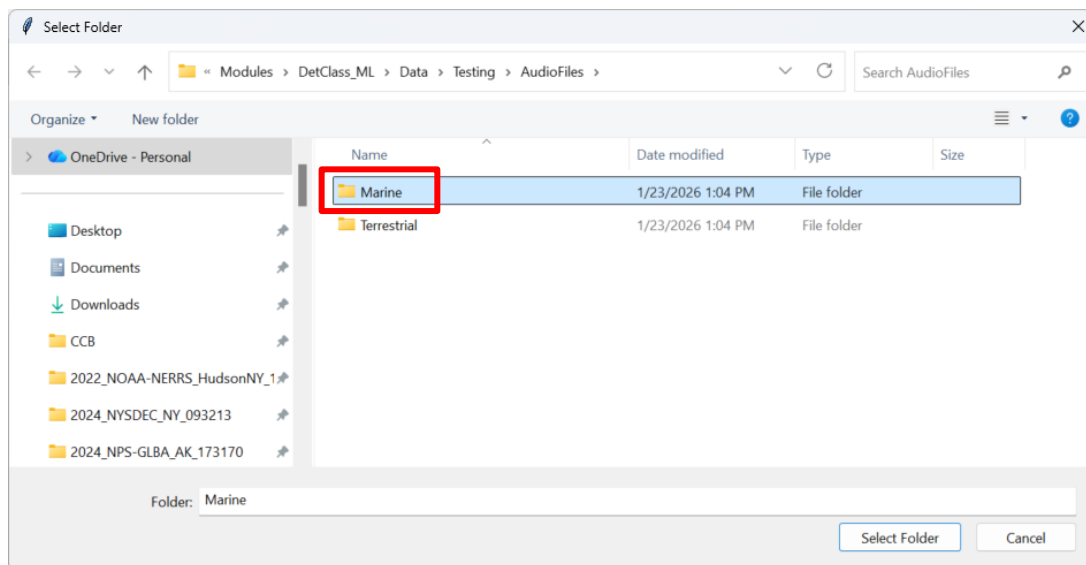
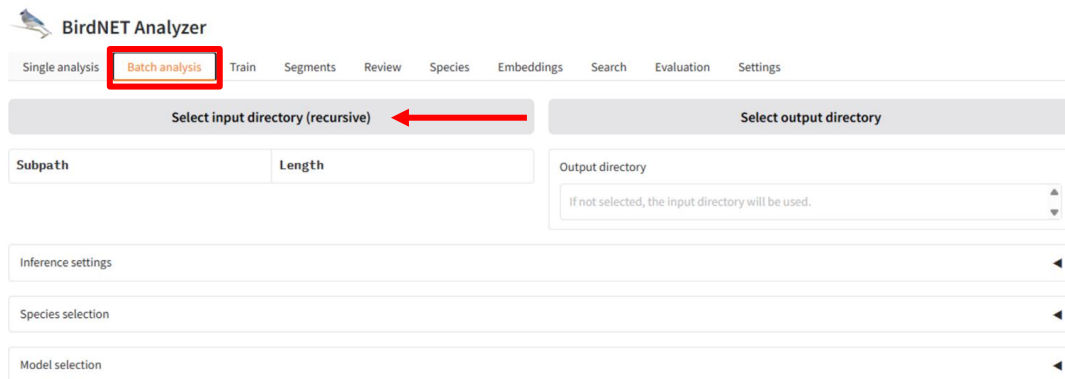
17.2s

Analyze

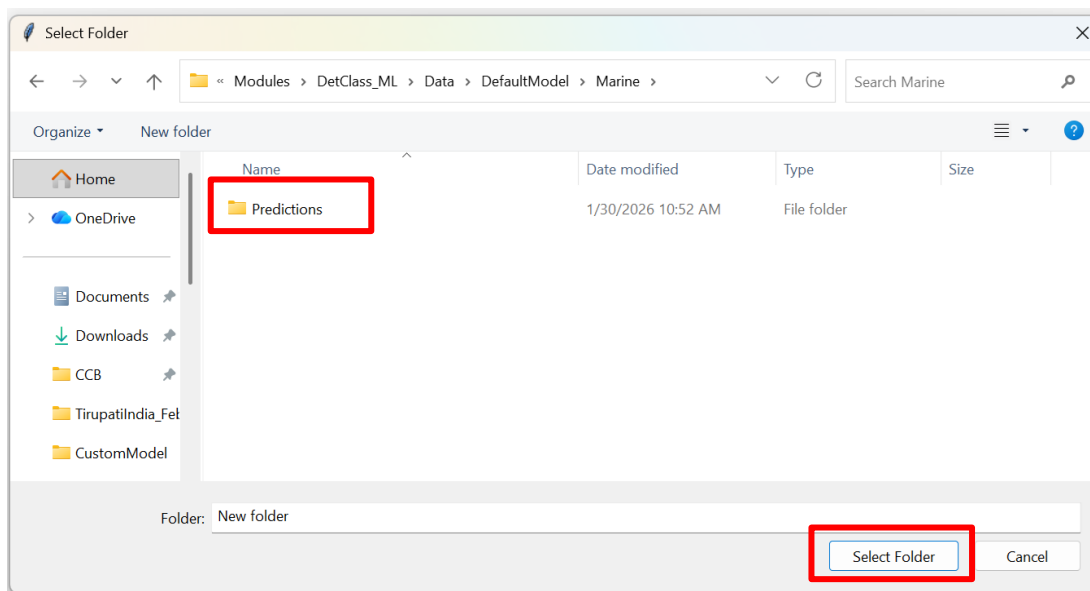
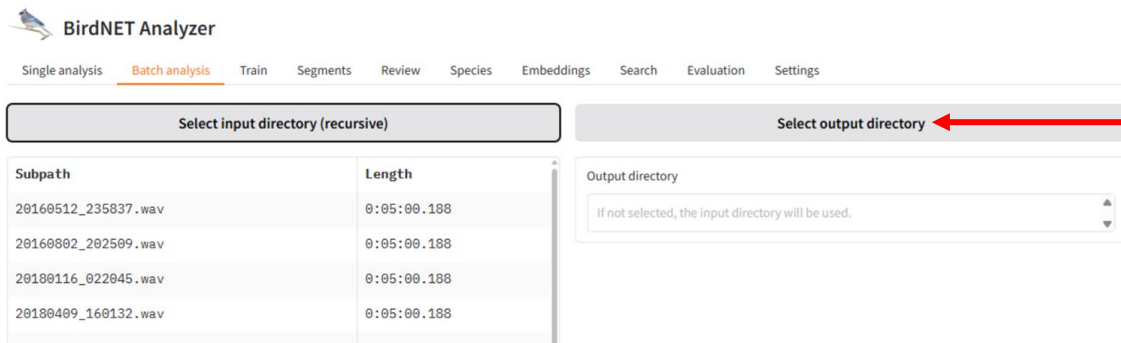
All files analyzed!

2.3 Running BirdNET Analyzer on Other Data

6. In the **Batch analysis** tab, click **Select input directory** and navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing >> Audio Files >> Marine*.



7. Click **Select output directory** and navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Marine >> Predictions*.



8. Under **Inference settings**, set **Maximum Bandpass Frequency (Hz)** to 10000
9. At the bottom of the screen, click **Analyze**. A progress bar will appear. When the analysis is complete it will switch to the message "All files analyzed!" Navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Marine >> Predictions* to see the output selection tables.

Output settings

Result type

Specify the output format of classifications.

☒ Raven selection table☐ Audacity☐ CSV☐ Kaleidoscope

Check this option to merge all selection tables into a single table.

☐ Combine selection tables

Skip files that already have a result.

☐ Skip existing results

Batch size

Number of samples to process at the same time.

1

Threads

Number of CPU threads.

4

Locale

Locale for the translated species common names in the output

EN

Analyze

Analyze

Starting ... - 0.0%

17.2s

Analyze

All files analyzed!

Predictions

DefaultModel > Marine > Predictions

Search Predictions

New

Sort

View

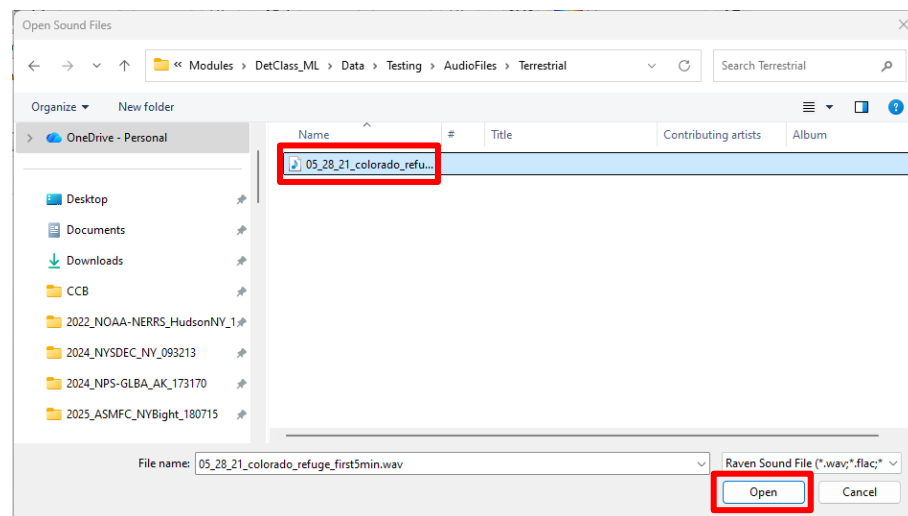
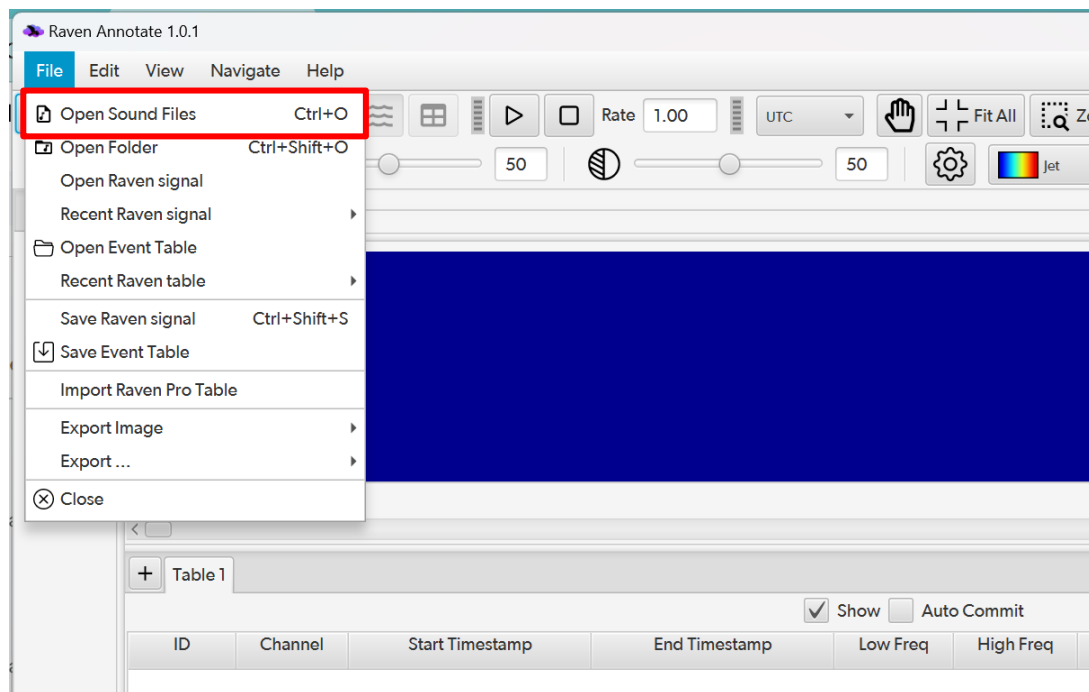
Details

Name	Date modified	Type	Size
20180116_022045.BirdNET.selection.table	1/30/2026 10:55 AM	Text Document	2 KB
20160802_202509.BirdNET.selection.table	1/30/2026 10:55 AM	Text Document	1 KB
20160512_235837.BirdNET.selection.table	1/30/2026 10:55 AM	Text Document	3 KB
20180409_160132.BirdNET.selection.table	1/30/2026 10:55 AM	Text Document	4 KB
20180629_145625.BirdNET.selection.table	1/30/2026 10:55 AM	Text Document	14 KB
20180710_000053.BirdNET.selection.table	1/30/2026 10:55 AM	Text Document	9 KB
20180822_121740.BirdNET.selection.table	1/30/2026 10:55 AM	Text Document	9 KB

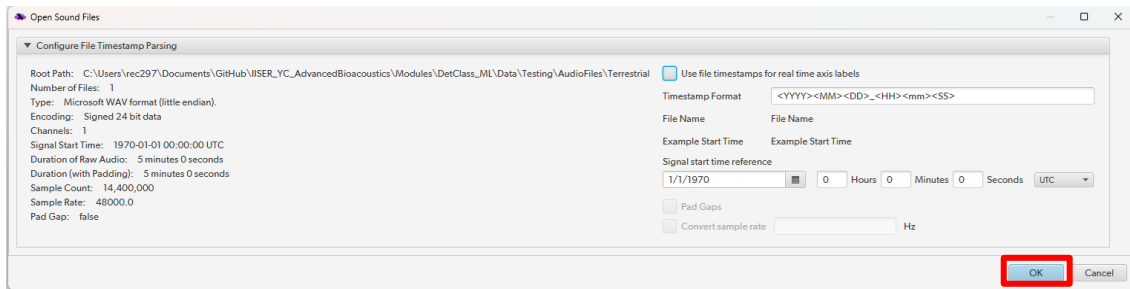
Part 3: Performance Evaluation

3.1 Viewing Annotations and Audio in Raven Annotate

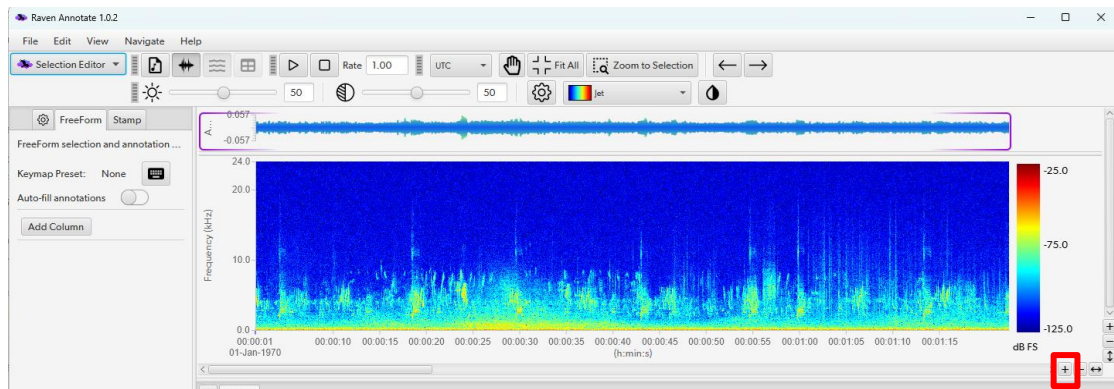
1. Launch Raven Annotate, Pro, or Lite. Select **File >> Open Sound Files**. Navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing >> Audio Files >> Terrestrial* and select the audio file 05_28_21_colorado_refuge_first5min.wav then click **Open**.



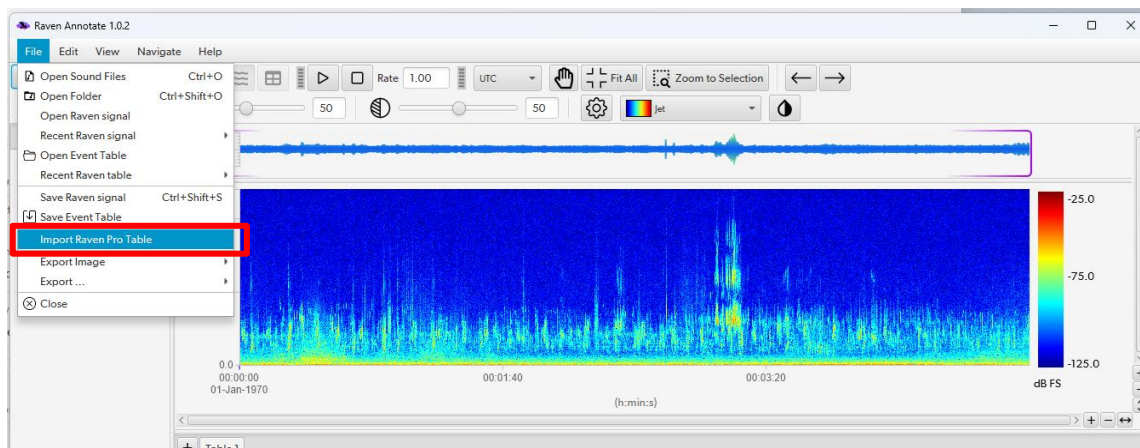
2. A dialogue box will appear. Uncheck the box for “Use file timestamps for real time axis labels”. Click **OK** and the data will load.

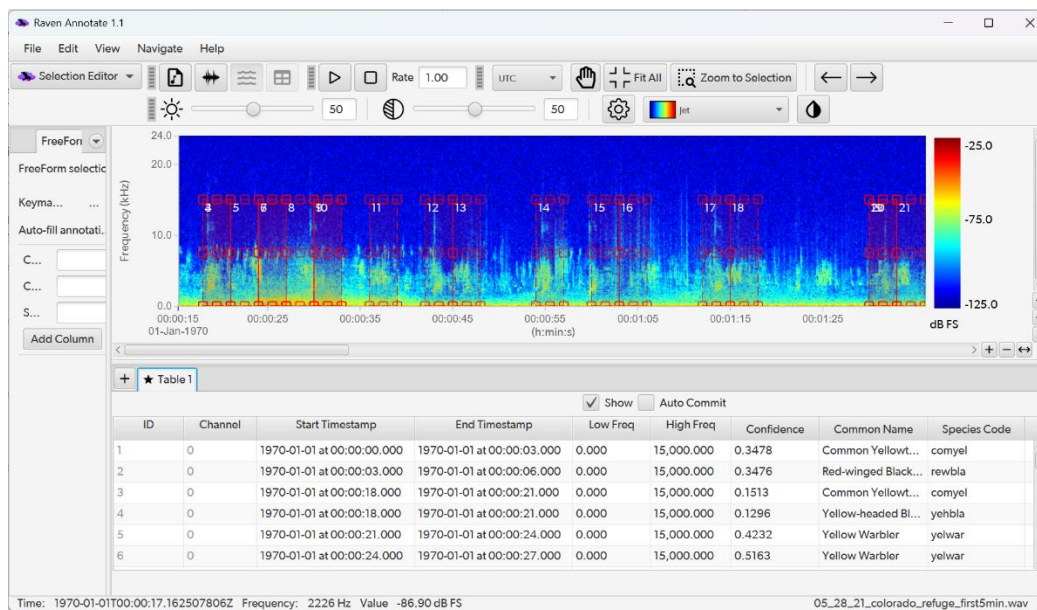
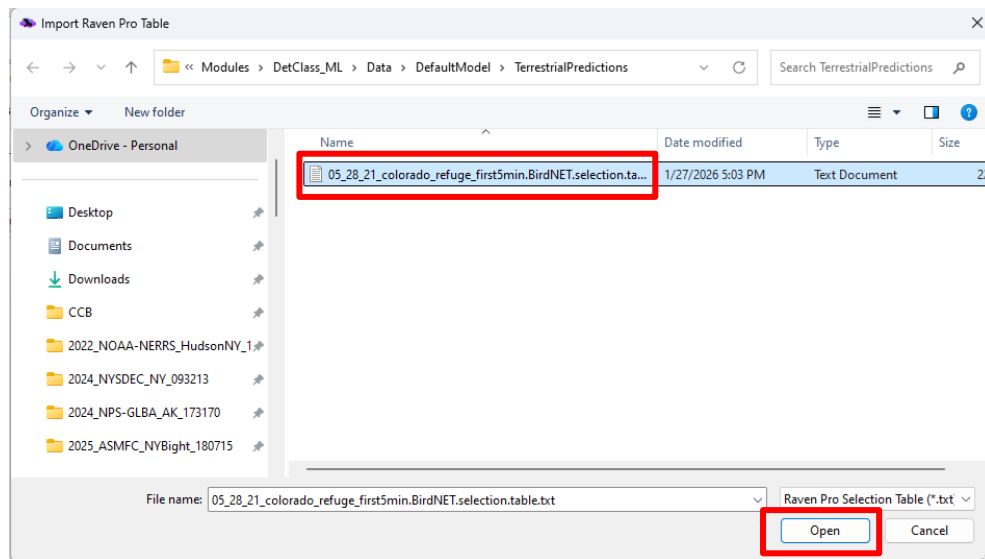


3. Zoom in on the x-axis by clicking the + in the bottom right corner of the spectrogram panel. Use the x-axis scroll bar to navigate to the start of the file.



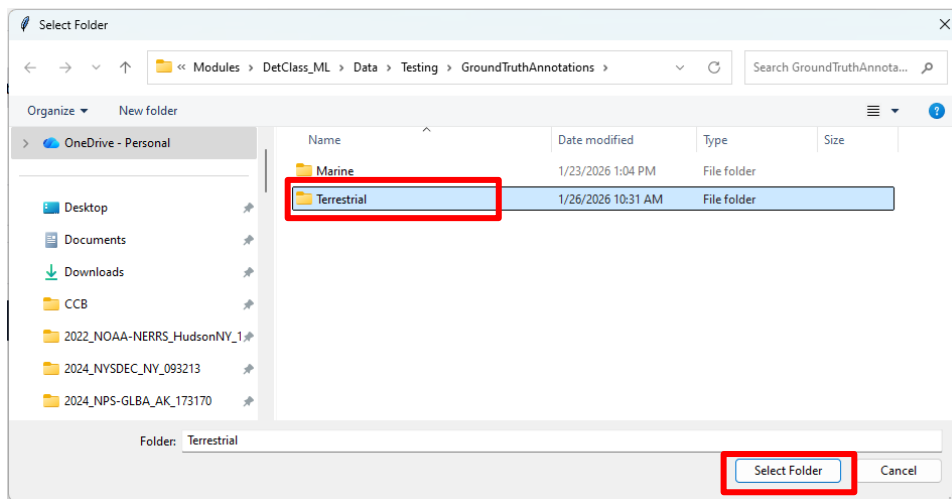
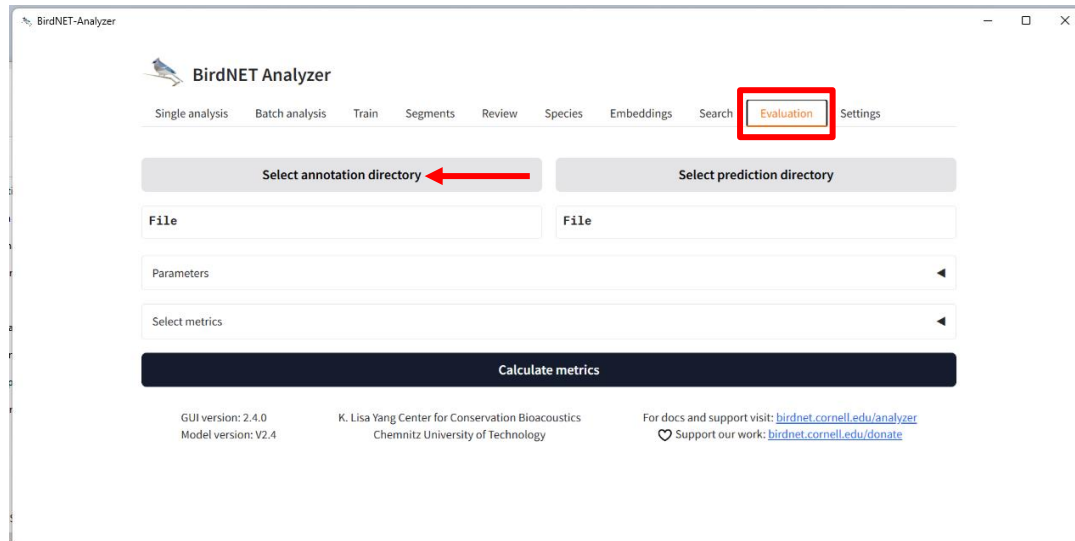
4. Select **File >> Import Raven Pro Table**. Navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel>> Terrestrial >> Predictions* and select *5_28_21_colorado_refuge_first5min.BirdNET.selection.table.txt*, then click **Open**.





3.2 Quantifying Model Performance with BirdNET

1. Click on the **Evaluation** tab. Click **Select annotation directory**, navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing >> GroundTruthAnnotations>>Terrestrial*, and click **Select Folder**.



2. Click **Select prediction directory**, navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Terrestrial >> Predictions*, and click **Select**

BirdNET Analyzer

Single analysis Batch analysis Train Segments Review Species Embeddings Search **Evaluation** Settings

Select annotation directory

Select prediction directory 

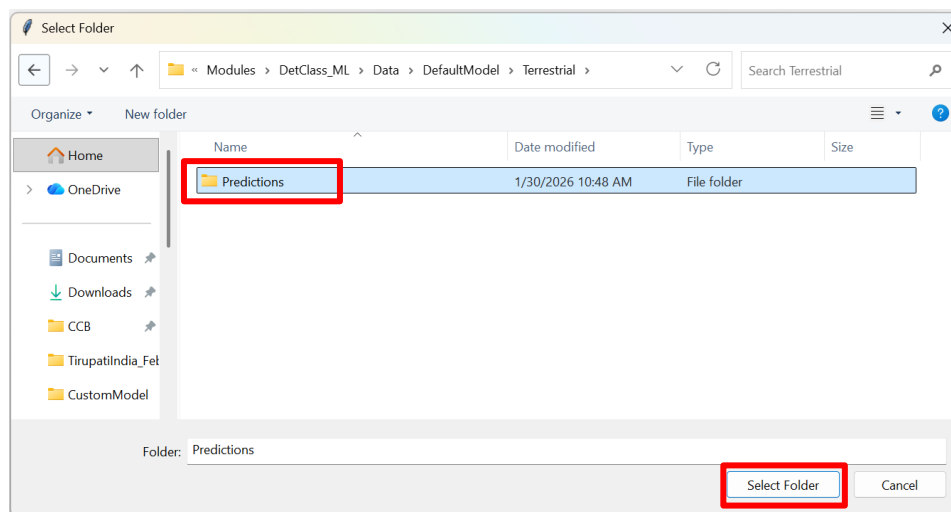
File

C:\Users\rec297\Documents\GitHub\IISER_YC_AdvancedBioacoustics\Modules

File

Annotation columns

Start time	End time	Class	Recording	Duration
------------	----------	-------	-----------	----------



3. Scroll down to **Annotation columns**. Under **Class**, use the drop down menu to select **Species**.

Annotation columns

Start time	End time	Class	Recording	Duration
Begin Time (s)	End Time (s)	Species	Begin File	

4. Click **Calculate metrics**. A results table with precision and recall will be generated. Plot precision/recall/F1/accuracy vs. threshold by clicking **Plot metrics for all thresholds**.

Calculate metrics

Plot metrics

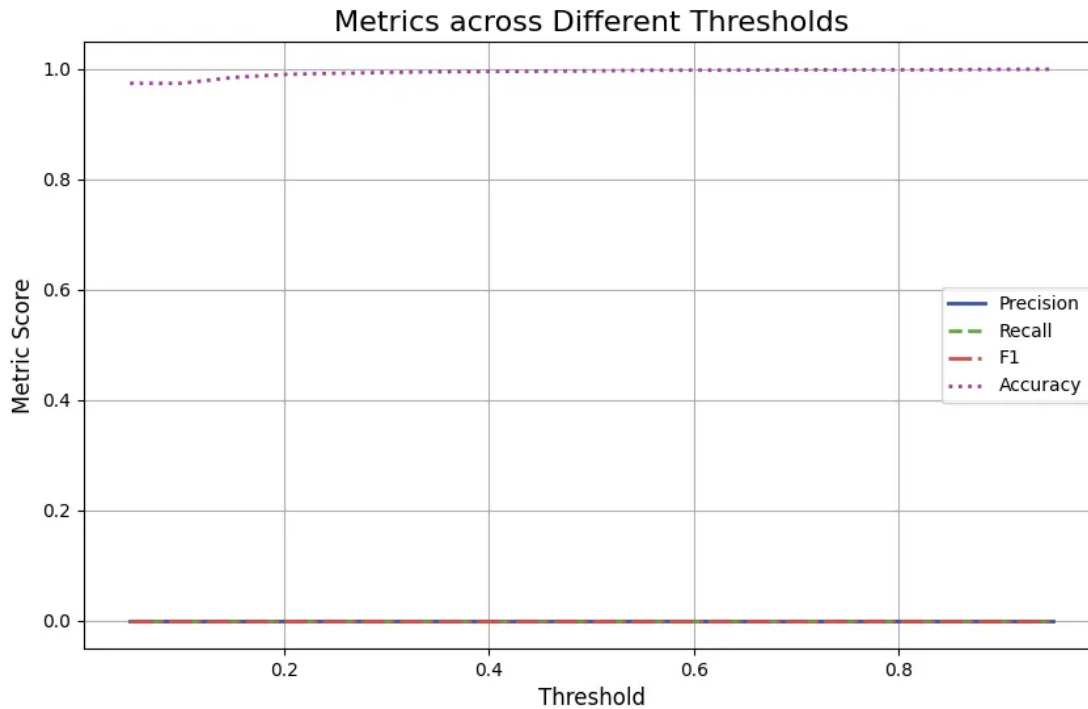
Plot confusion matrix

Plot metrics for all thresholds

Download results table

Download data table

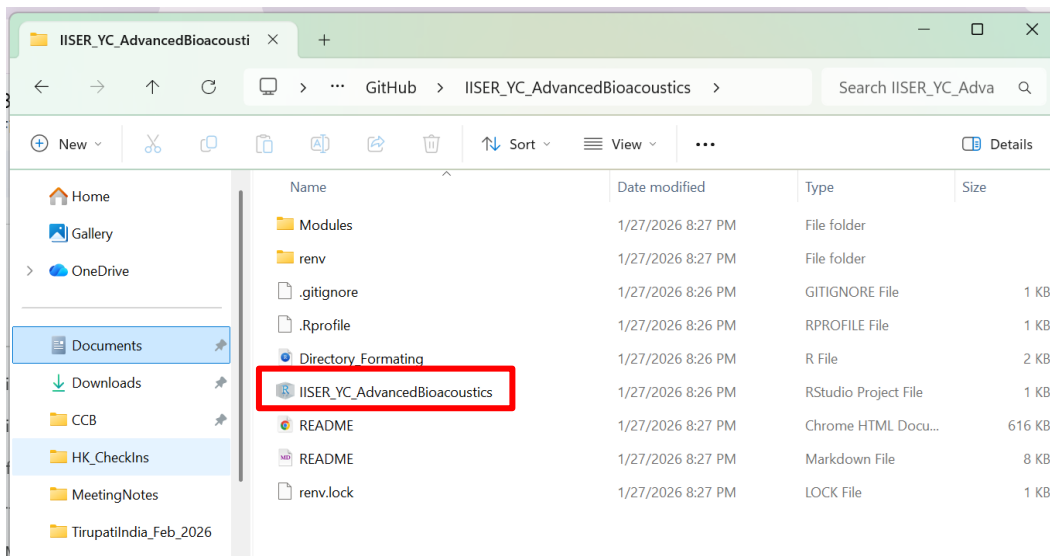
	AUROC	Precision	Recall	F1	AP	Accuracy
Overall	null	0	0	0	0	0.9736111111111112



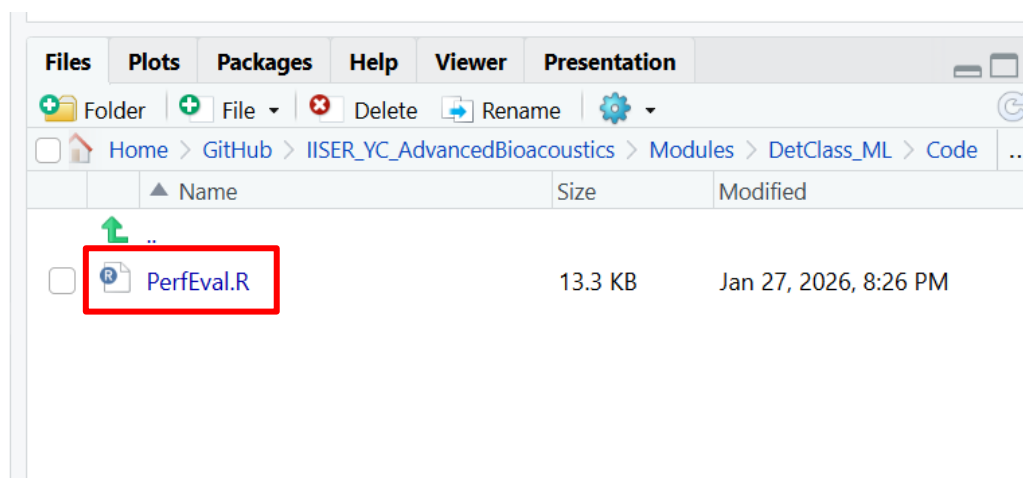
- Repeat process with marine ground truth (*IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing >> GroundTruthAnnotations>>Marine*) and marine BirdNET predictions (*IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Marine >> Predictions*).

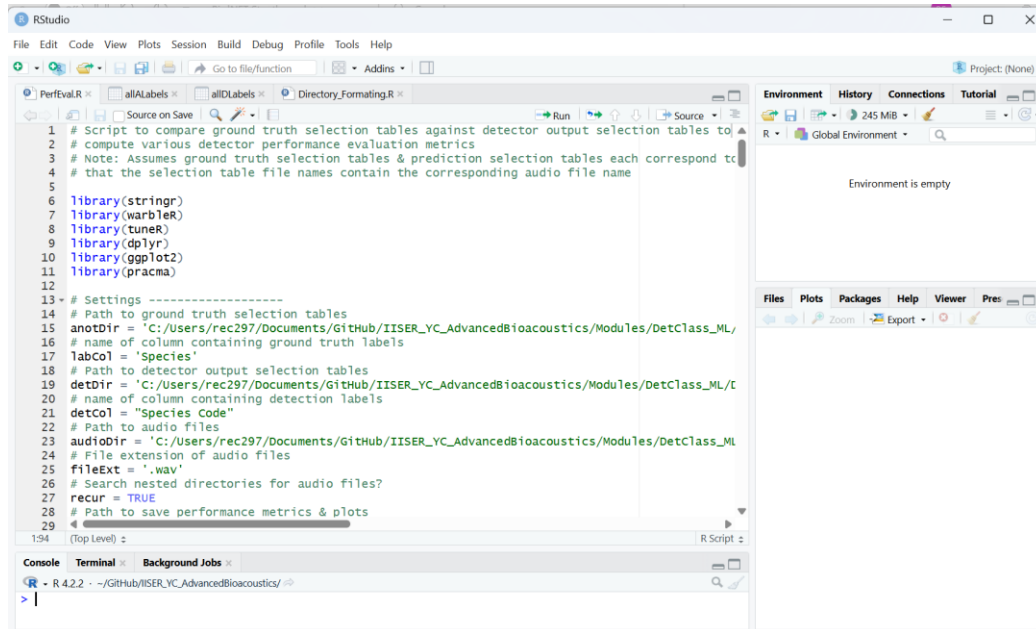
3.3 Quantifying Model Performance with Custom Scripting

1. Launch the workshop R Project by navigating to the downloaded GitHub repository folder *IISER_YC_AdvancedBioacoustics* and double clicking the file *IISER_YC_AdvancedBioacoustics.Rproj*.

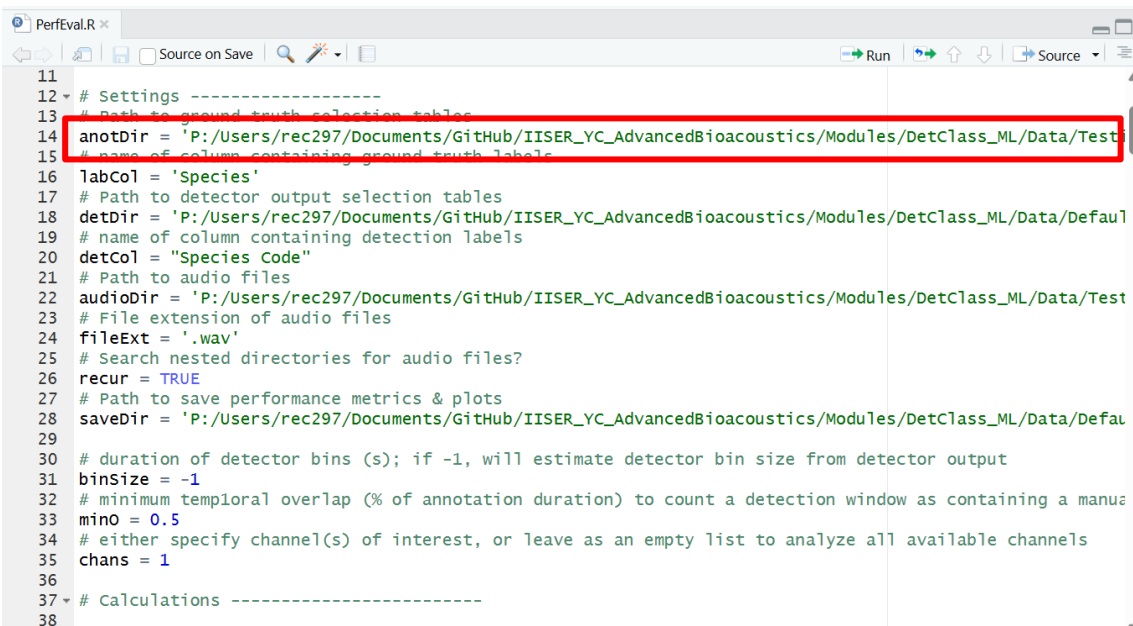


2. In the **Console** enter the command `renv::restore()`
3. In the **Files** pane of the RStudio session, navigate to *IISER_YC_AdvancedBioacoustics* >> *Modules* >> *DetClass_ML* >> *Code* and click *PerfEval.R* to open the script.

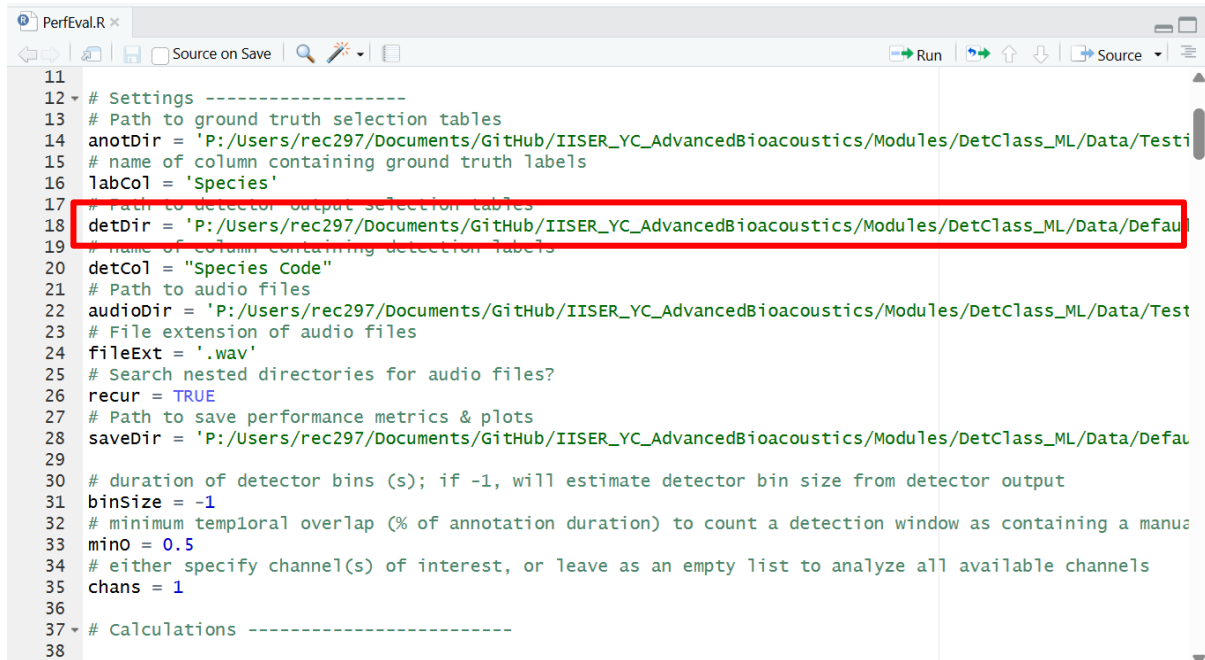




4. Update the `anotDir` variable path in line 14 to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing >> GroundTruthAnnotations >> Terrestrial*

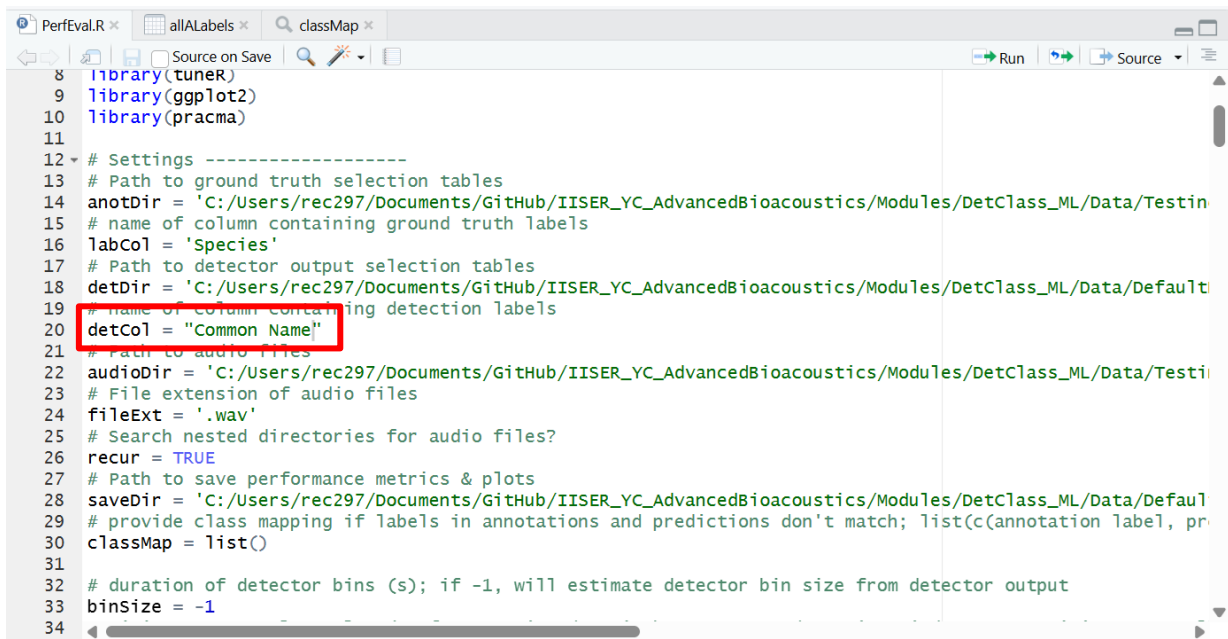


5. Update the `detDir` variable path in line 18 to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Terrestrial >> Predictions*



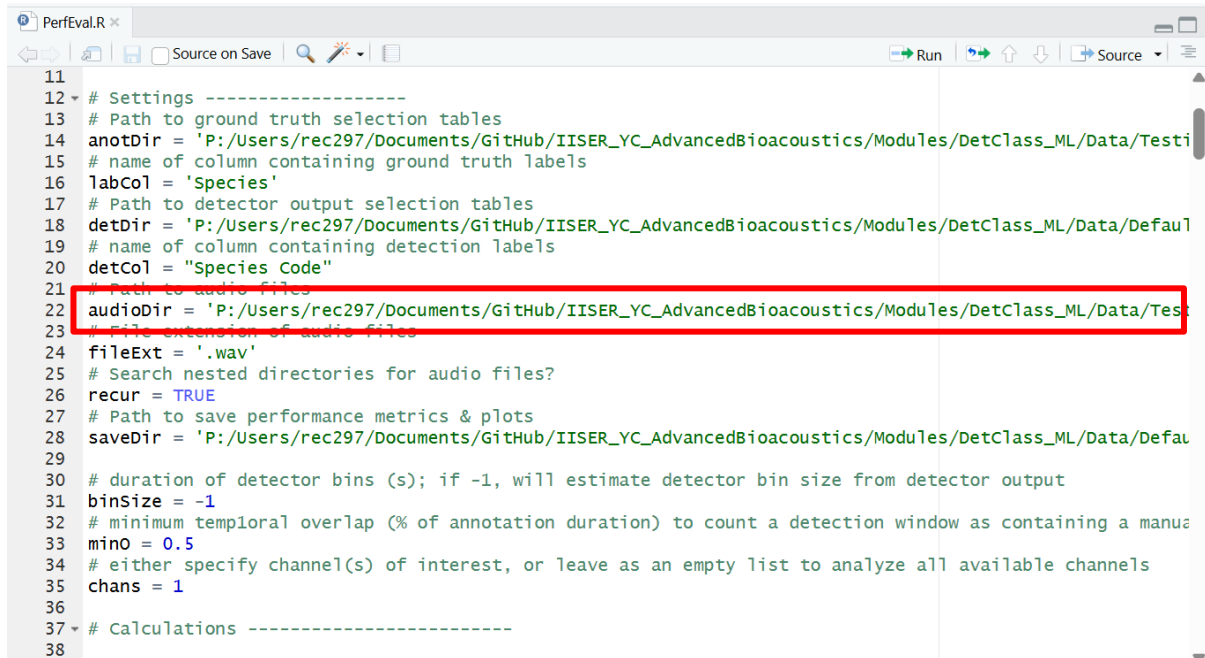
```
11
12 # Settings -----
13 # Path to ground truth selection tables
14 anotDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testi
15 # name of column containing ground truth labels
16 labCol = 'Species'
17 # Path to detector output selection tables
18 detDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Defau
19 # name of column containing detection labels
20 detCol = "Species Code"
21 # Path to audio files
22 audioDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testi
23 # File extension of audio files
24 fileExt = '.wav'
25 # Search nested directories for audio files?
26 recur = TRUE
27 # Path to save performance metrics & plots
28 saveDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Defau
29
30 # duration of detector bins (s); if -1, will estimate detector bin size from detector output
31 binSize = -1
32 # minimum temporal overlap (% of annotation duration) to count a detection window as containing a manual
33 minO = 0.5
34 # either specify channel(s) of interest, or leave as an empty list to analyze all available channels
35 chans = 1
36
37 # Calculations -----
38
```

6. Update the `detCol` variable in line 20 to “Common Name”



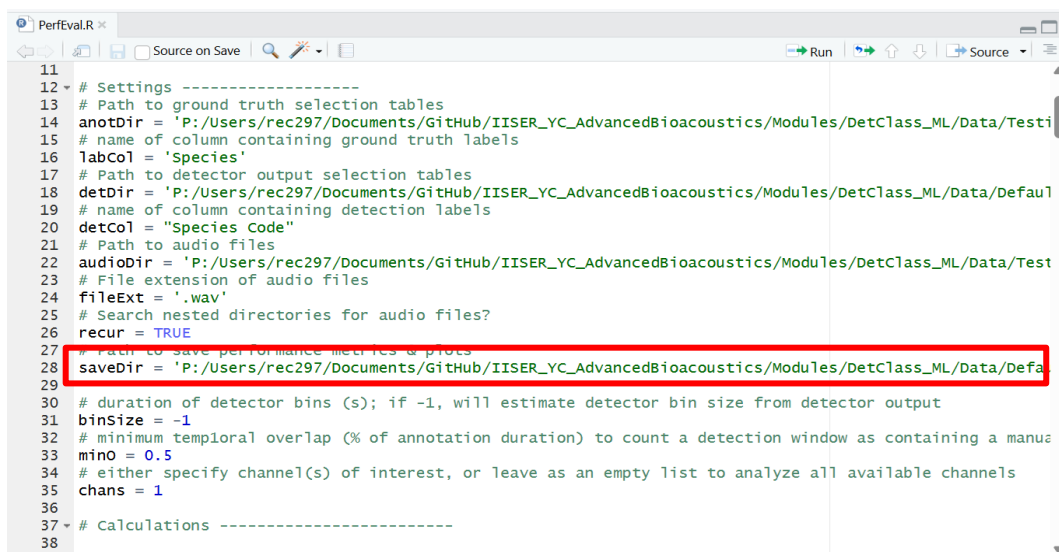
```
8 library(tuner)
9 library(ggplot2)
10 library(pracma)
11
12 # Settings -----
13 # Path to ground truth selection tables
14 anotDir = 'C:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testin
15 # name of column containing ground truth labels
16 labCol = 'Species'
17 # Path to detector output selection tables
18 detDir = 'C:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Default
19 # name of column containing detection labels
20 detCol = "Common Name"
21 # Path to audio files
22 audioDir = 'C:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testi
23 # File extension of audio files
24 fileExt = '.wav'
25 # Search nested directories for audio files?
26 recur = TRUE
27 # Path to save performance metrics & plots
28 saveDir = 'C:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Defaul
29 # provide class mapping if labels in annotations and predictions don't match; list(c(annotation label, pr
30 classMap = list()
31
32 # duration of detector bins (s); if -1, will estimate detector bin size from detector output
33 binSize = -1
34
```

7. Update the `audioDir` variable path in line 22 to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing >> AudioFiles >> Terrestrial*



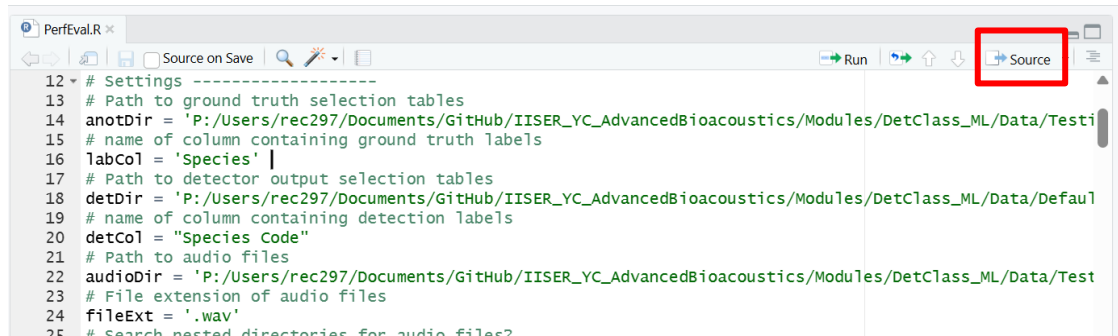
```
11
12 # Settings -----
13 # Path to ground truth selection tables
14 anotDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testing/AudioFiles/Terrestrial'
15 # name of column containing ground truth labels
16 labCol = 'Species'
17 # Path to detector output selection tables
18 detDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/DefaultModel/Terrestrial'
19 # name of column containing detection labels
20 detCol = "Species Code"
21 # Path to audio files
22 audioDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testing/AudioFiles/Terrestrial'
23 # File extension of audio files
24 fileExt = '.wav'
25 # Search nested directories for audio files?
26 recur = TRUE
27 # Path to save performance metrics & plots
28 savedir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/DefaultModel/Terrestrial'
29
30 # duration of detector bins (s); if -1, will estimate detector bin size from detector output
31 binSize = -1
32 # minimum temporal overlap (% of annotation duration) to count a detection window as containing a manual annotation
33 minO = 0.5
34 # either specify channel(s) of interest, or leave as an empty list to analyze all available channels
35 chans = 1
36
37 # Calculations -----
38
```

8. Update the `savedir` variable path in line 28 to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Terrestrial*



```
11
12 # Settings -----
13 # Path to ground truth selection tables
14 anotDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testing/AudioFiles/Terrestrial'
15 # name of column containing ground truth labels
16 labCol = 'Species'
17 # Path to detector output selection tables
18 detDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/DefaultModel/Terrestrial'
19 # name of column containing detection labels
20 detCol = "Species Code"
21 # Path to audio files
22 audioDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testing/AudioFiles/Terrestrial'
23 # File extension of audio files
24 fileExt = '.wav'
25 # Search nested directories for audio files?
26 recur = TRUE
27 # Path to save performance metrics & plots
28 savedir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/DefaultModel/Terrestrial'
29
30 # duration of detector bins (s); if -1, will estimate detector bin size from detector output
31 binSize = -1
32 # minimum temporal overlap (% of annotation duration) to count a detection window as containing a manual annotation
33 minO = 0.5
34 # either specify channel(s) of interest, or leave as an empty list to analyze all available channels
35 chans = 1
36
37 # Calculations -----
38
```

9. Run script by clicking **Source** button in upper right corner of script editor window



```
12 # Settings -----
13 # Path to ground truth selection tables
14 anotDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testi
15 # name of column containing ground truth labels
16 labCol = 'Species' |
17 # Path to detector output selection tables
18 detDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Defau1
19 # name of column containing detection labels
20 detCol = "Species Code"
21 # Path to audio files
22 audioDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Test
23 # File extension of audio files
24 fileExt = '.wav'
25 # Search nested directories for audio files?
```

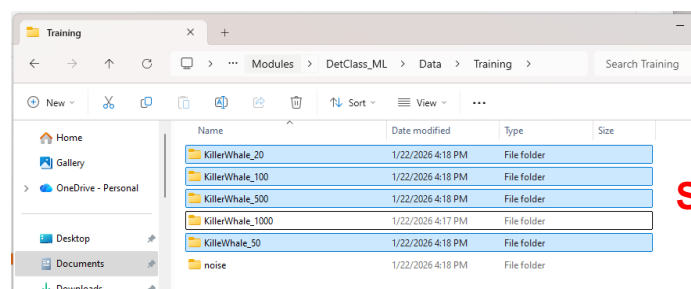
10. In Windows Explorer or Finder, navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> DefaultModel >> Terrestria >> IPredictions* to view output figures.
11. Repeat process for marine data predictions.

Part 4: Custom BirdNET Models

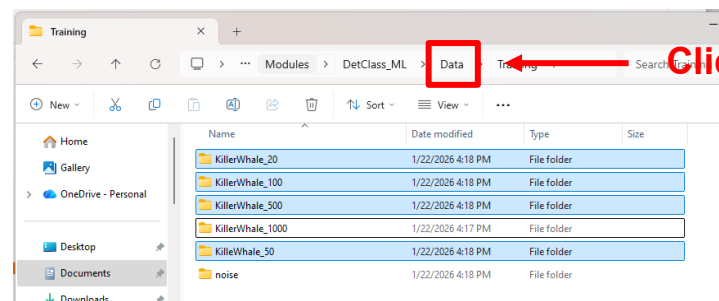
We can use BirdNET to detect sound classes that it was not originally trained on by applying transfer learning with a modest amount of training data. To train a custom BirdNET model, you need a training dataset consisting of short sound clips from both the target class(es) (specific species or sound type(s)) and at least one non-target class (e.g. background noise).

4.1 Training a Custom Model

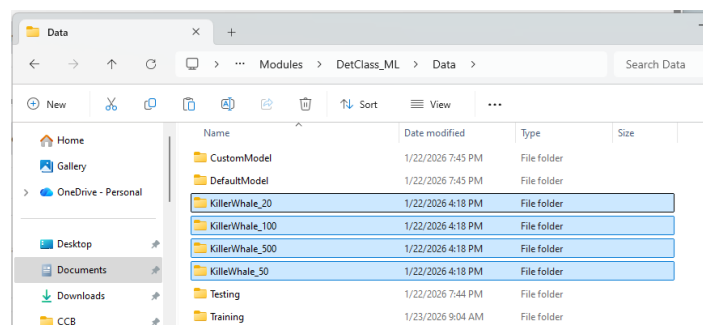
1. We will train a custom model with two classes: Killer Whale and Noise. BirdNET will use all training data folders in the directory you point it towards, so we'll need to reorganize the data slightly so that each team uses a different number of training examples. In the workshop folder, navigate to *Modules >> DetClass_ML >> Data >> Training*. **Select and cut all killer whale training data folders except the one assigned to your group. Paste them up one level, directly into the Data folder.**



Select, Ctrl+X

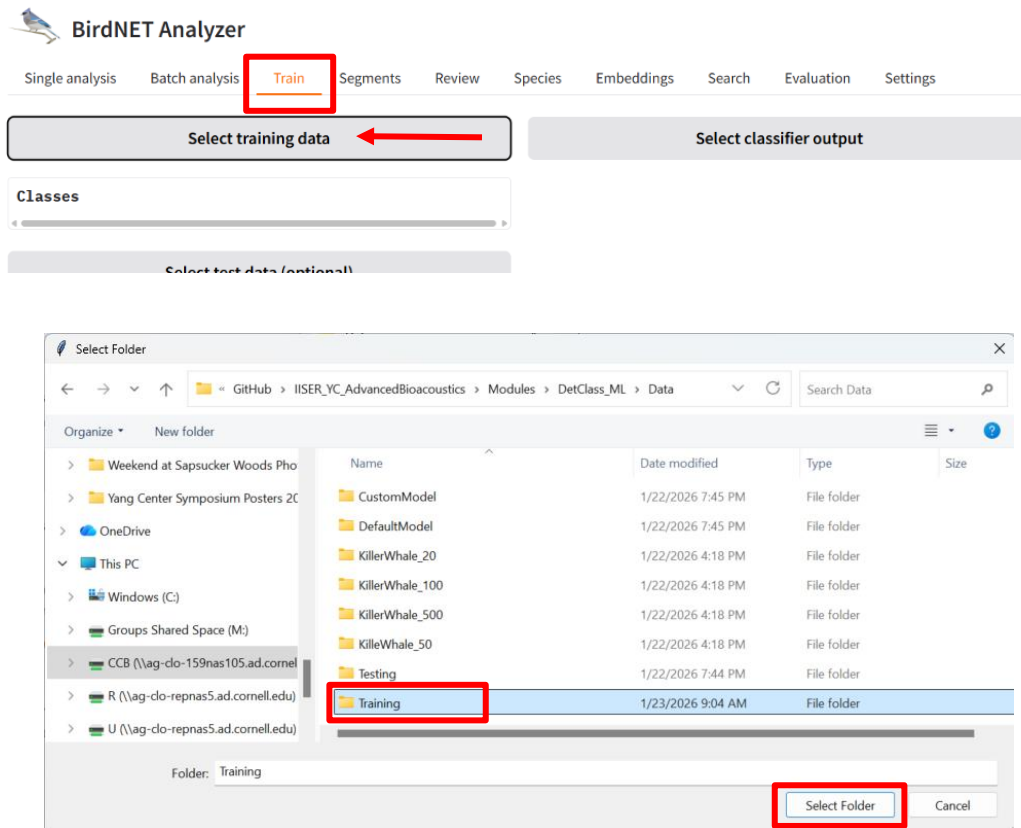


Click

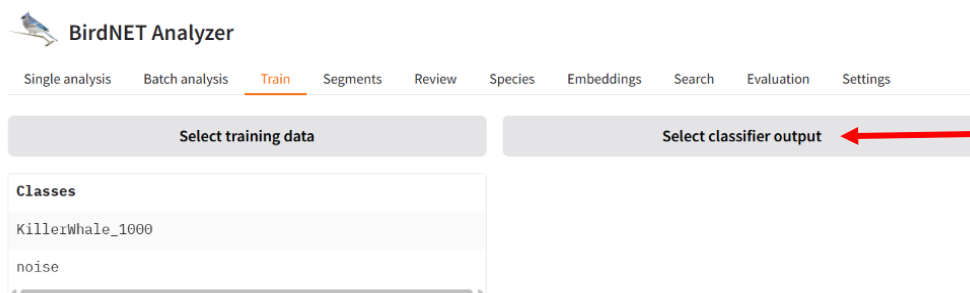


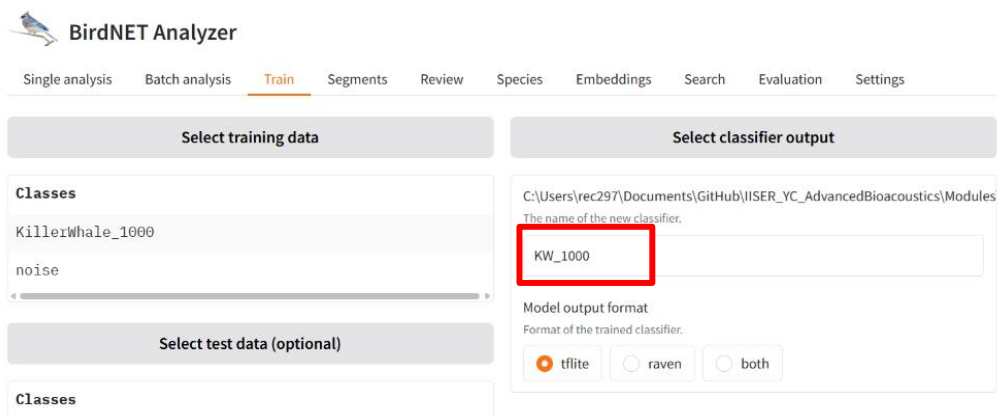
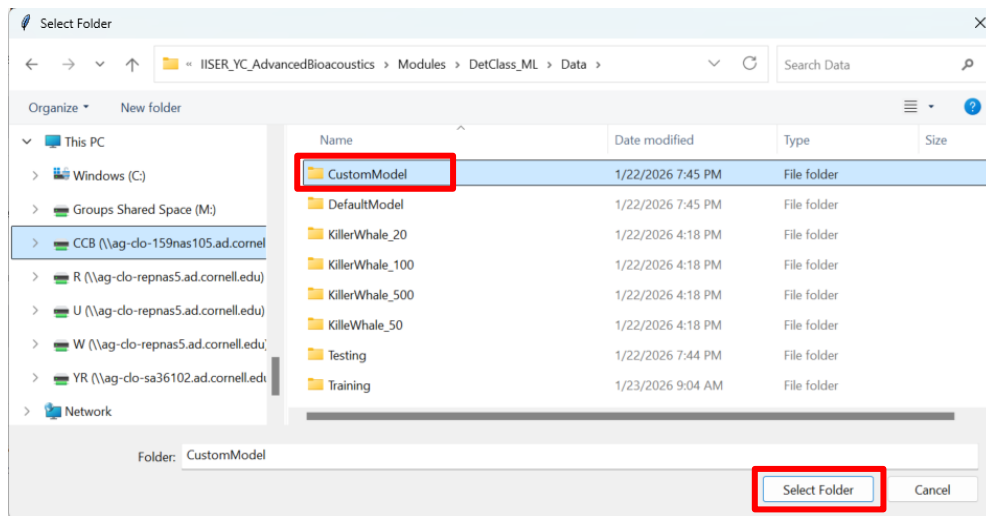
Ctrl+V

2. In **BirdNET Analyzer**, navigate to the **Train** tab. Click **Select training data**, navigate to *Modules >> DetClass_ML >> Data >> Training*, and click **Select**.

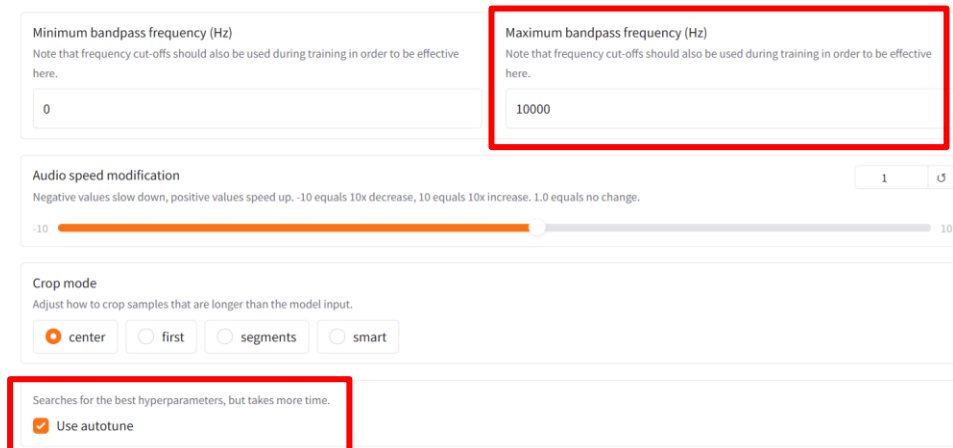


3. Click **Select classifier output**, navigate to *Modules >> DetClass_ML >> Data >> CustomModel*, and click **Select**. You can also specify a name for your custom model (by default it will be called "CustomClassifier").

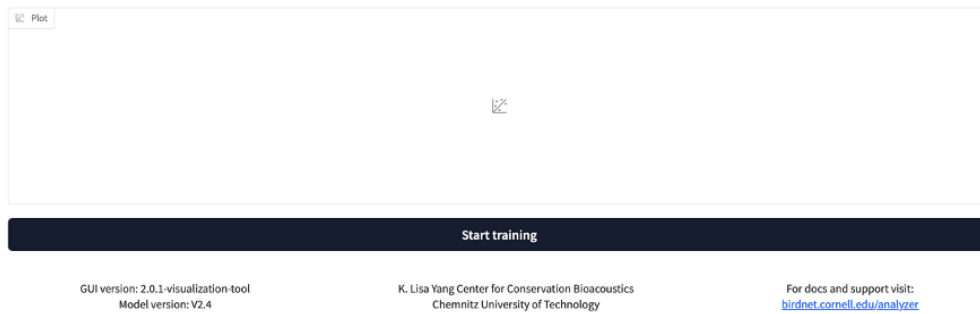




- Set the **Maximum bandpass frequency (Hz)** to 10000. Click the box next to **Use autotune** to implement automatic hyperparameter optimization.

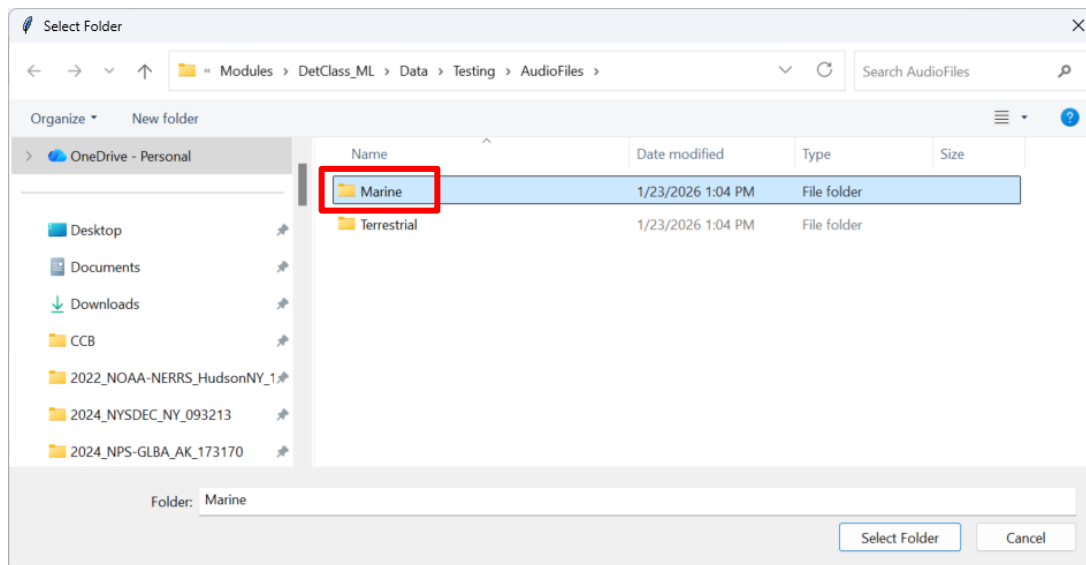
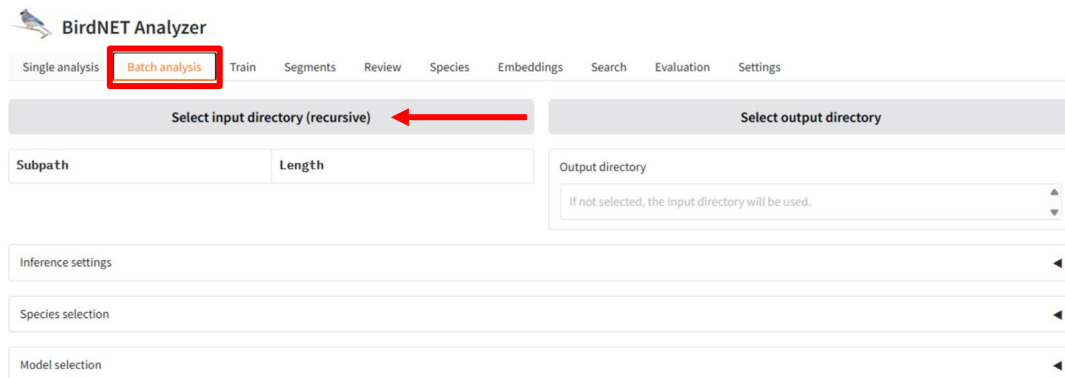


5. Scroll to the bottom of the train tab and click ***Start training***.

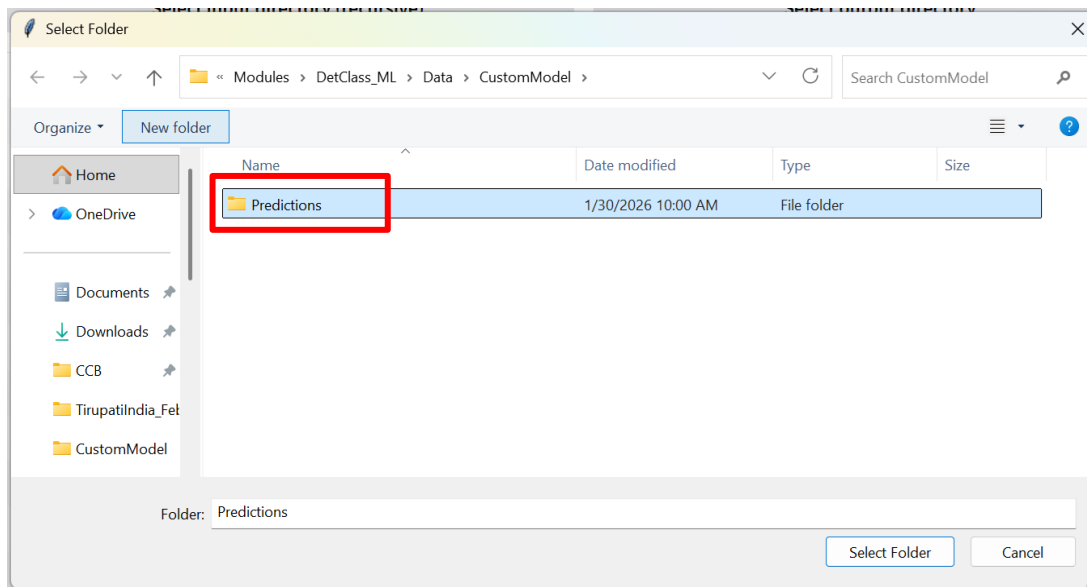
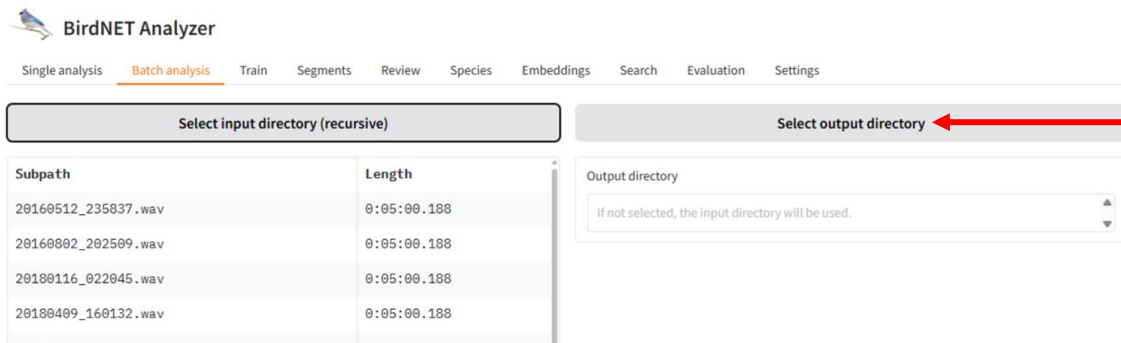


4.2 Deploying a Custom Model

1. In the **Batch analysis** tab, click **Select input directory** and navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing >> Audio Files >> Marine*.



2. Click **Select output directory** and navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> CustomModel*, select *Predictions*, and click **Select folder**.



- Under **Inference settings**, set **Maximum Bandpass Frequency (Hz)** to 10000

Inference settings

Picks the top N species sorted by confidence score.

☐ Use top N species

Sensitivity

Adjust the distribution of prediction scores. Higher values result in higher scores.

1

↕

0.5 ————— 1.5

Merge consecutive detections

Number of consecutive detections that are merged for each species. Combines N consecutive detections into one.

1

↕

1 ————— 10

Minimum bandpass frequency (Hz)

Note that frequency cut-offs should also be used during training in order to be effective here.

0

Minimum confidence

Adjust the threshold to ignore results with confidence below this level.

0.25

↕

0.05 ————— 0.95

Overlap (s)

BirdNET uses 3s segments. Determines the overlap with previous segment.

0

↕

0 ————— 2.9

Audio speed modification

Negative values slow down, positive values speed up. Note that audio speed modification should also be used during training in order to be effective here.

1

↕

-10 ————— 10

Maximum bandpass frequency (Hz)

Note that frequency cut-offs should also be used during training in order to be effective here.

10000

- Under **Model selection**, select **Custom classifier**, then click the **Select classifier** button and navigate to **Modules >> DetClass_ML >> Data >> CustomModel**, select the classifier you just trained and click **Open**.

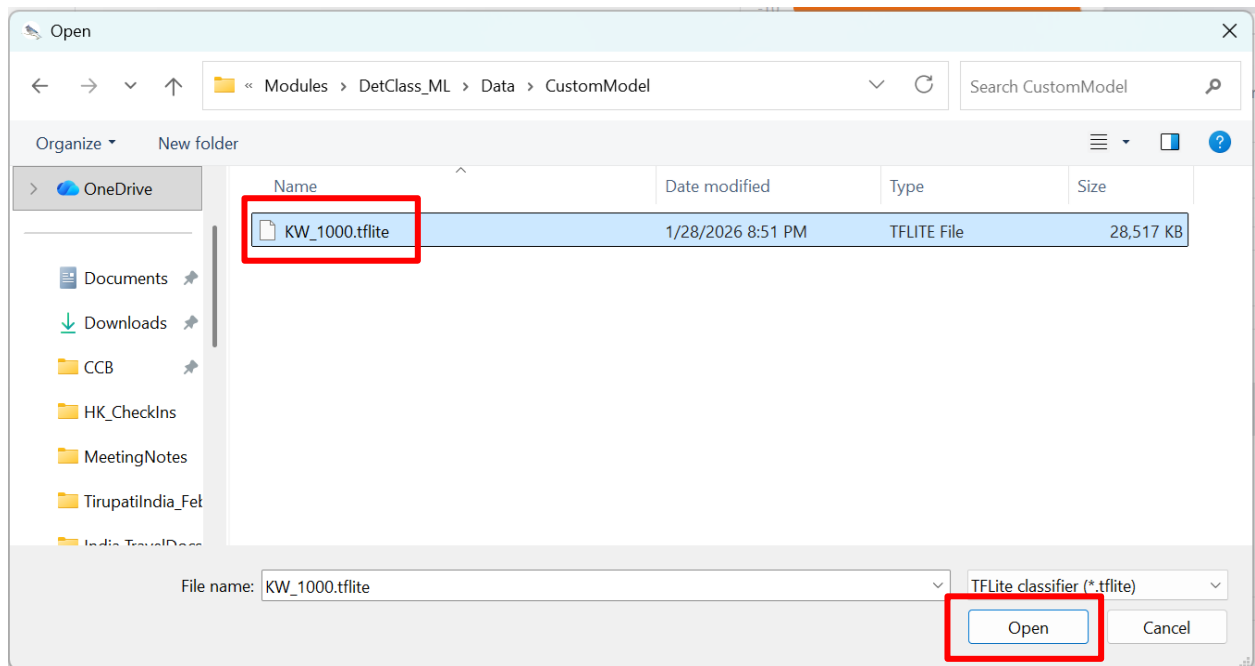
Model selection

Select the model

The selected model will be used for analysis.

☐ BirdNET V2.4
☒ Custom classifier
☐ Perch v2 (preview)

Select classifier



5. Scroll to the bottom of the window and click **Analyze** to deploy the custom model.

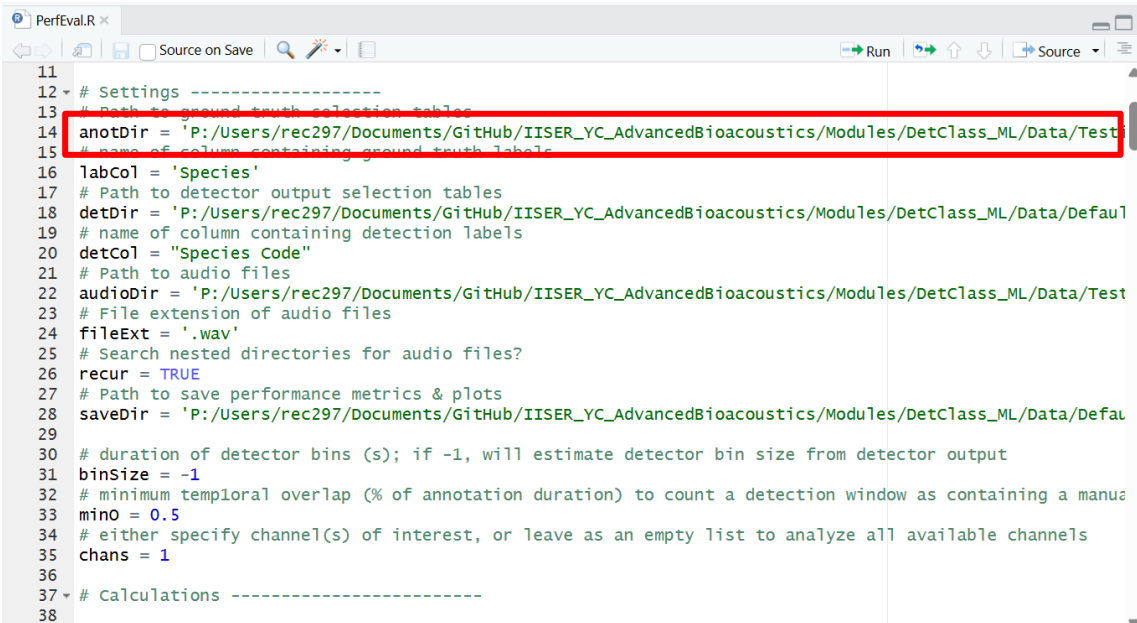
Number of samples to process (minimum: 1): Number of epochs:

Locale
Locale for the translated species common names in the output

Analyze ←

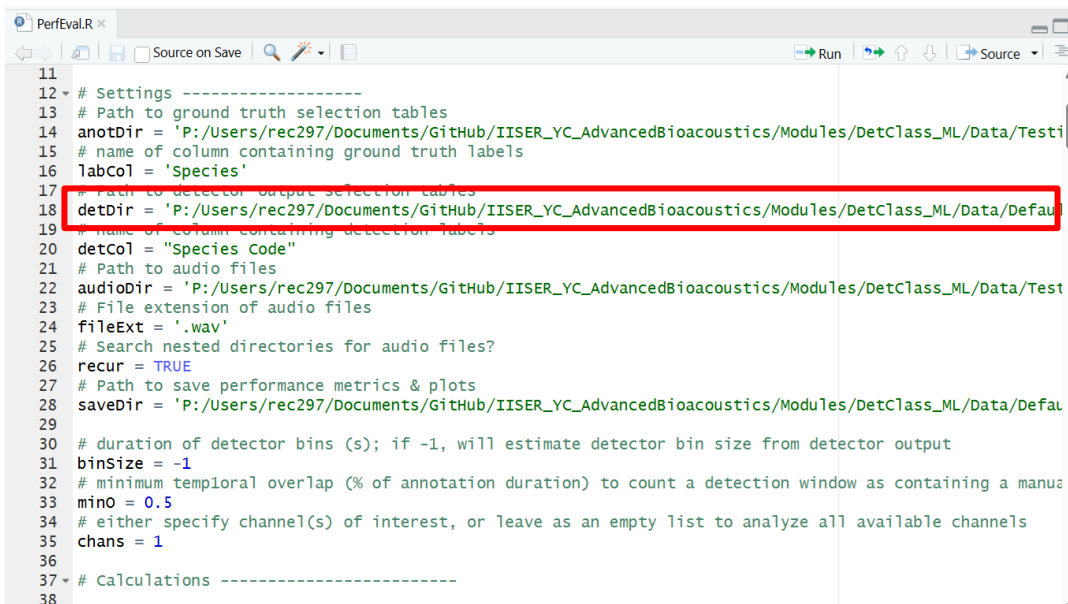
4.3 Evaluating Custom Model Performance

1. Update the `anotDir` variable path in line 14 to *IISER_YC_AdvancedBioacoustics*
>> *Modules* >> *DetClass_ML* >> *Data* >> *Testing* >> *GroundTruthAnnotations* >> *Marine*



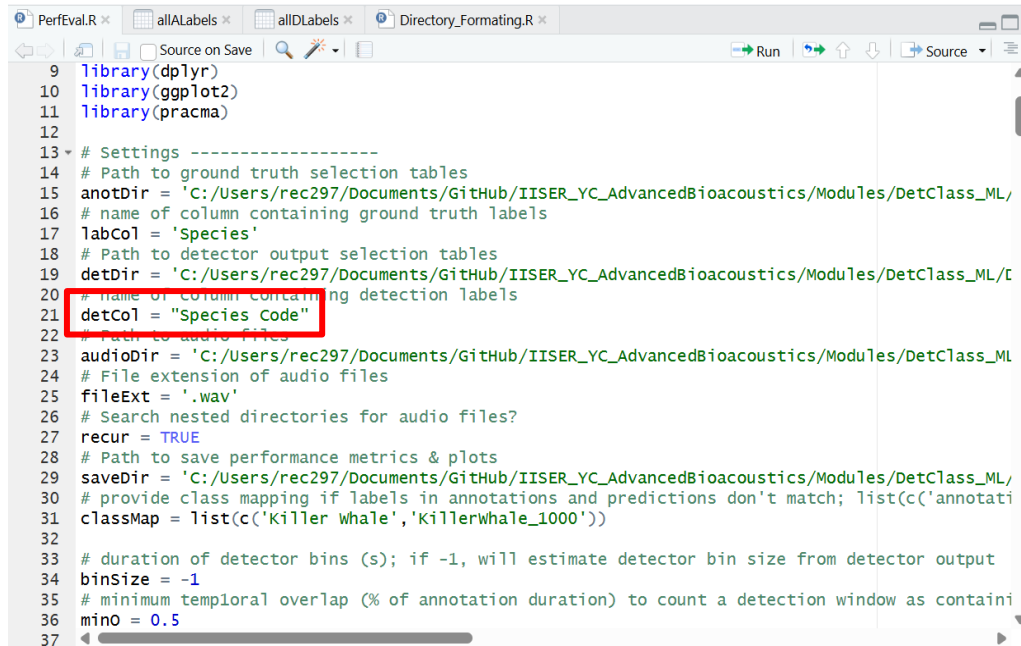
```
11
12 # Settings -----
13 # Path to ground truth selection tables
14 anotDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testing/GroundTruthAnnotations/Marine'
15 # name of column containing ground truth labels
16 labCol = 'Species'
17 # Path to detector output selection tables
18 detDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/DefaultPredictions'
19 # name of column containing detection labels
20 detCol = "Species Code"
21 # Path to audio files
22 audioDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testing/AudioFiles'
23 # File extension of audio files
24 fileExt = '.wav'
25 # Search nested directories for audio files?
26 recur = TRUE
27 # Path to save performance metrics & plots
28 saveDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/DefaultMetrics'
29
30 # duration of detector bins (s); if -1, will estimate detector bin size from detector output
31 binSize = -1
32 # minimum temporal overlap (% of annotation duration) to count a detection window as containing a manual annotation
33 minO = 0.5
34 # either specify channel(s) of interest, or leave as an empty list to analyze all available channels
35 chans = 1
36
37 # Calculations -----
38
```

2. Update the `detDir` variable path in line 18 to *IISER_YC_AdvancedBioacoustics*
>> *Modules* >> *DetClass_ML* >> *Data* >> *CustomModel* >> *Predictions*



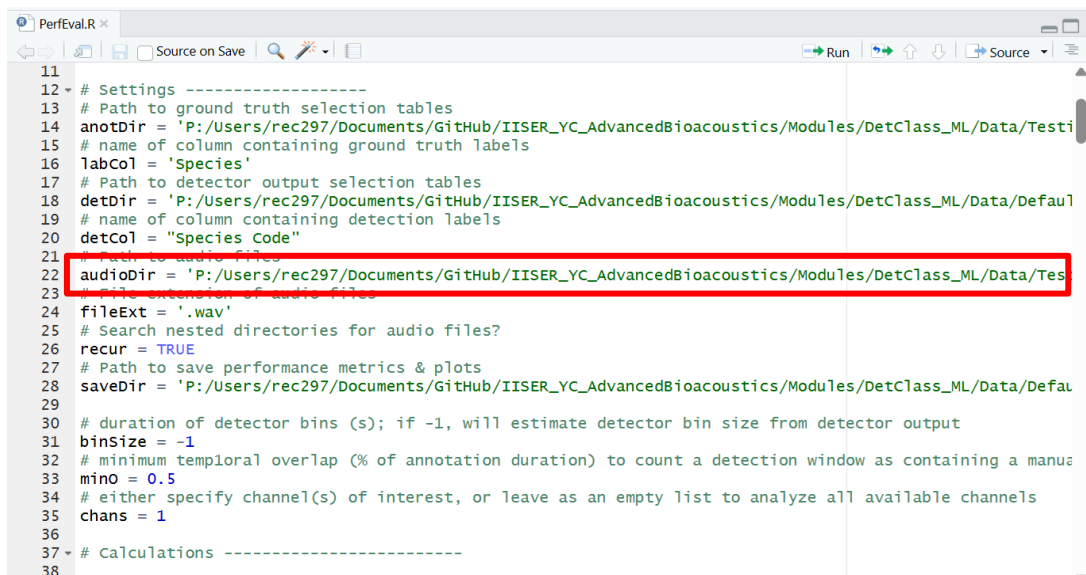
```
11
12 # Settings -----
13 # Path to ground truth selection tables
14 anotDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testing/GroundTruthAnnotations/Marine'
15 # name of column containing ground truth labels
16 labCol = 'Species'
17 # Path to detector output selection tables
18 detDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/DefaultPredictions'
19 # name of column containing detection labels
20 detCol = "Species Code"
21 # Path to audio files
22 audioDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testing/AudioFiles'
23 # File extension of audio files
24 fileExt = '.wav'
25 # Search nested directories for audio files?
26 recur = TRUE
27 # Path to save performance metrics & plots
28 saveDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/DefaultMetrics'
29
30 # duration of detector bins (s); if -1, will estimate detector bin size from detector output
31 binSize = -1
32 # minimum temporal overlap (% of annotation duration) to count a detection window as containing a manual annotation
33 minO = 0.5
34 # either specify channel(s) of interest, or leave as an empty list to analyze all available channels
35 chans = 1
36
37 # Calculations -----
38
```

3. Update the detCol variable in line 20 to "Species Code"



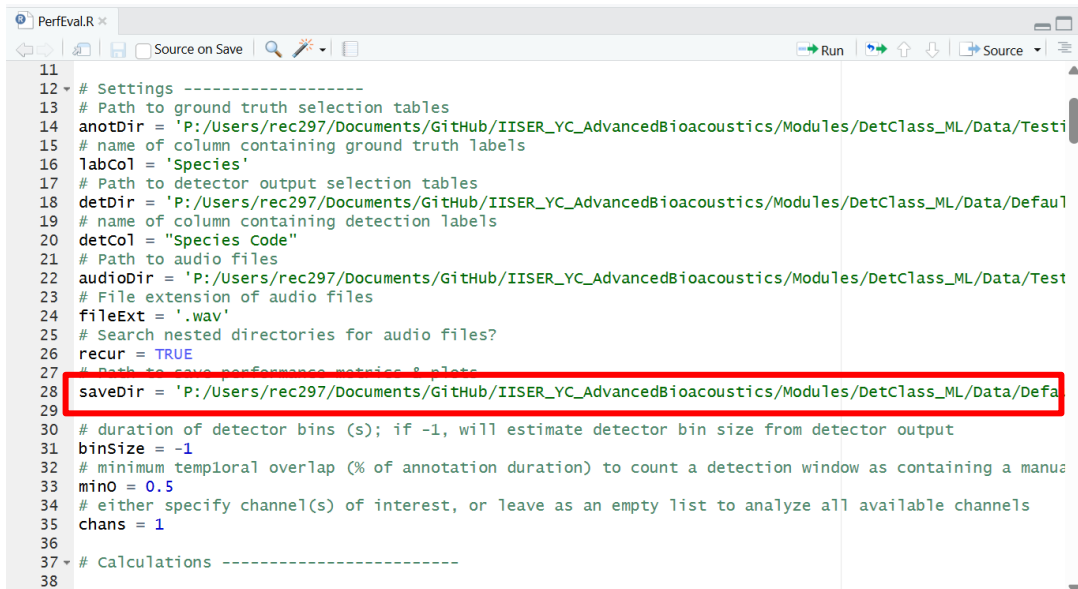
```
9 library(dplyr)
10 library(ggplot2)
11 library(pracma)
12
13 # Settings -----
14 # Path to ground truth selection tables
15 anotDir = 'C:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/
16 # name of column containing ground truth labels
17 labCol = 'Species'
18 # Path to detector output selection tables
19 detDir = 'C:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/I
20 # name of column containing detection labels
21 detCol = "Species Code"
22 # Path to audio files
23 audioDir = 'C:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML
24 # File extension of audio files
25 fileExt = '.wav'
26 # Search nested directories for audio files?
27 recur = TRUE
28 # Path to save performance metrics & plots
29 saveDir = 'C:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/
30 # provide class mapping if labels in annotations and predictions don't match; list(c('annotati
31 classMap = list(c('Killer Whale','KillerWhale_1000'))
32
33 # duration of detector bins (s); if -1, will estimate detector bin size from detector output
34 binSize = -1
35 # minimum temporal overlap (% of annotation duration) to count a detection window as containi
36 minO = 0.5
37
```

4. Update the audioDir variable path in line 22 to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> Testing>> AudioFiles >> Marine*



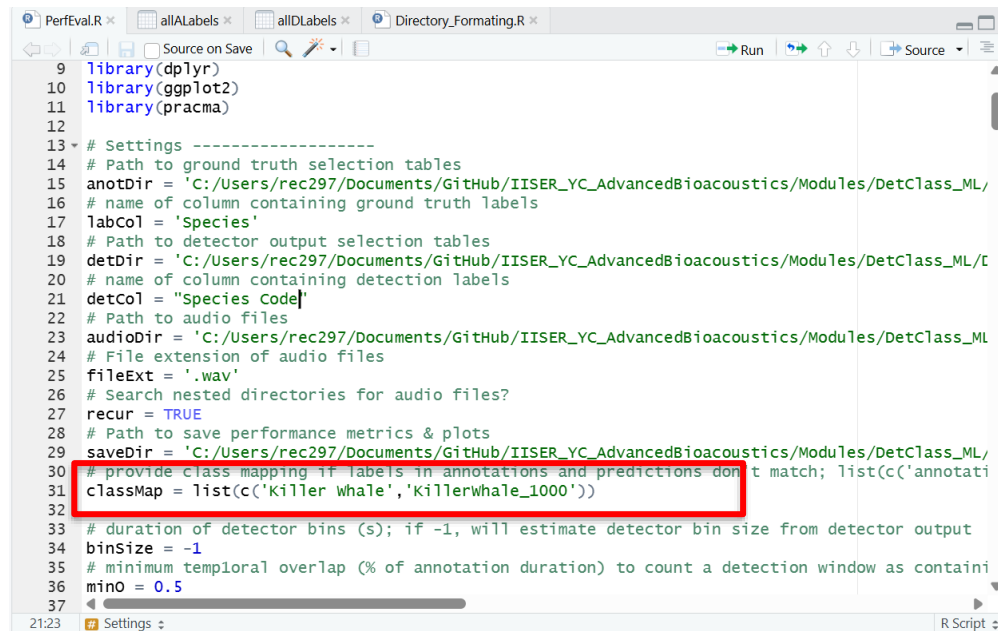
```
11
12 # Settings -----
13 # Path to ground truth selection tables
14 anotDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testing
15 # name of column containing ground truth labels
16 labCol = 'Species'
17 # Path to detector output selection tables
18 detDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Default
19 # name of column containing detection labels
20 detCol = "Species Code"
21 # Path to audio files
22 audioDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testing/AudioFiles/Marine'
23 # File extension of audio files
24 fileExt = '.wav'
25 # Search nested directories for audio files?
26 recur = TRUE
27 # Path to save performance metrics & plots
28 saveDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Default
29
30 # duration of detector bins (s); if -1, will estimate detector bin size from detector output
31 binSize = -1
32 # minimum temporal overlap (% of annotation duration) to count a detection window as containing a manual
33 minO = 0.5
34 # either specify channel(s) of interest, or leave as an empty list to analyze all available channels
35 chans = 1
36
37 # Calculations -----
38
```

5. Update the `saveDir` variable path in line 28 to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> CustomModel*



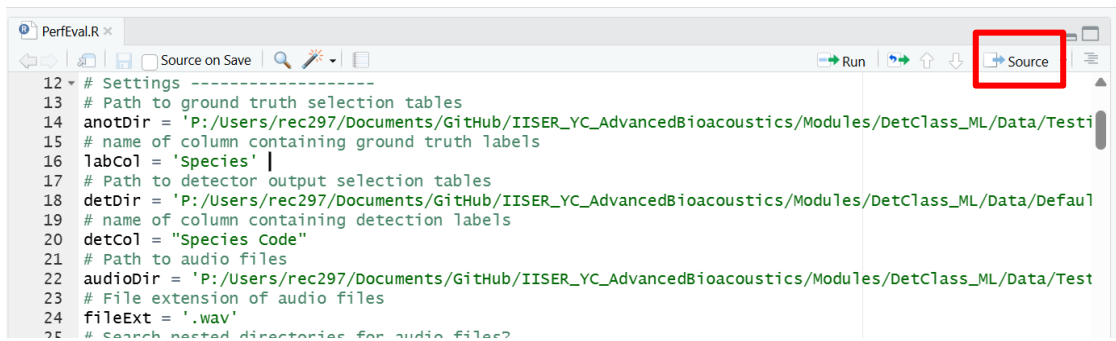
```
11
12 # Settings -----
13 # Path to ground truth selection tables
14 anotDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testi
15 # name of column containing ground truth labels
16 labCol = 'Species'
17 # Path to detector output selection tables
18 detDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Defaul
19 # name of column containing detection labels
20 detCol = "Species Code"
21 # Path to audio files
22 audioDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Test
23 # File extension of audio files
24 fileExt = '.wav'
25 # Search nested directories for audio files?
26 recur = TRUE
27 # Path to save performance metrics & plots
28 saveDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Defa
29
30 # duration of detector bins (s); if -1, will estimate detector bin size from detector output
31 binSize = -1
32 # minimum temporal overlap (% of annotation duration) to count a detection window as containing a manus
33 mino = 0.5
34 # either specify channel(s) of interest, or leave as an empty list to analyze all available channels
35 chans = 1
36
37 # Calculations -----
38
```

6. Update `classMap` variable in line 31 to reflect name of training data folder your group used.



```
9 library(dplyr)
10 library(ggplot2)
11 library(pracma)
12
13 # Settings -----
14 # Path to ground truth selection tables
15 anotDir = 'C:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/
16 # name of column containing ground truth labels
17 labCol = 'Species'
18 # Path to detector output selection tables
19 detDir = 'C:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/C
20 # name of column containing detection labels
21 detCol = "Species Code"
22 # Path to audio files
23 audioDir = 'C:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML
24 # File extension of audio files
25 fileExt = '.wav'
26 # Search nested directories for audio files?
27 recur = TRUE
28 # Path to save performance metrics & plots
29 saveDir = 'C:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/
30 # provide class mapping if labels in annotations and predictions don't match; list(c('annotati
31 classMap = list(c('Killer whale', 'Killerwhale_1000'))
32
33 # duration of detector bins (s); if -1, will estimate detector bin size from detector output
34 binSize = -1
35 # minimum temporal overlap (% of annotation duration) to count a detection window as containi
36 mino = 0.5
37
21:23 Settings R Script
```

7. Run script by clicking **Source** button in upper right corner of script editor window



The screenshot shows the RStudio interface with a script editor window titled 'PerfEval.R'. The script contains R code for setting paths and variables. The 'Source' button in the top right corner of the editor is highlighted with a red rectangle. The code in the editor is as follows:

```
12 # Settings -----
13 # Path to ground truth selection tables
14 anotDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Testi
15 # name of column containing ground truth labels
16 labCol = 'Species'
17 # Path to detector output selection tables
18 detDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Defau1
19 # name of column containing detection labels
20 detCol = "Species Code"
21 # Path to audio files
22 audioDir = 'P:/Users/rec297/Documents/GitHub/IISER_YC_AdvancedBioacoustics/Modules/DetClass_ML/Data/Test
23 # File extension of audio files
24 fileExt = '.wav'
25 # Search nested directories for audio files?
```

8. In Windows Explorer or Finder, navigate to *IISER_YC_AdvancedBioacoustics >> Modules >> DetClass_ML >> Data >> CustomModel* to view output figures.